INTERNATIONAL MONETARY FUND

From Calibration to Estimation: A Modern Bayesian Framework for Quarterly Projection Models

Prepared by Andrés González Gómez and Juan David Prada

WP/26/241

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2026
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IMF Working Paper
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Authorized for distribution by Rafael Portillo
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JEL Classification Numbers:	C11, C32, E31, E37, E52, F41
Keywords:	Bayesian estimation, Hamiltonian Monte Carlo, quarterly projection model, semi-structural models, monetary policy, Brazil, sacrifice ratio, natural interest rate
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* Thanks to Rafael Portillo, Jesper Lindé, Pau Rabanal, Tugrul Vehbi, Zoltán Jakab, and Mátyás Farkas for very useful comments.

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Abstract

Quarterly Projection Models (QPMs) are widely used at central banks but rarely estimated by full Bayesian methods, because rational-expectations solvers and shock-variance priors resist gradient-based inference. We develop a workflow pairing the structure-preserving doubling algorithm, automatic differentiation and the No-U-Turn Sampler with a signal-to-noise-ratio prior calibration refined by a Koopman disturbance-smoother EM step. Applied to an open-economy QPM for Brazil (2004Q1–2025Q2), it delivers stable posterior inference, tightens monetary-policy impulse-response bands relative to the prior, places the output gap inside the Banco Central do Brasil's published envelope in about 84% of the sample, and produces stable pseudo-real-time forecasts, with its clearest edge over a Bayesian VAR on the policy rate.

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‡**Acknowledgments:** Thanks to Rafael Portillo, Jesper Lindé, Pau Rabanal, Tugrul Vehbi, Zoltán Jakab, and Mátyás Farkas for very useful comments. Despite their input, we are responsible for any omissions and errors. The estimation code is available at https://github.com/agonzago/brazil_qpm_replication and is implemented in Python/JAX.

1 Introduction

Quarterly projection models (QPMs) are central to monetary policy analysis. Dozens of central banks use them to combine New Keynesian theory (IS curves, Phillips curves, Taylor rules, and UIP conditions) with judgment and conditional forecasts (Berg et al., 2006; Beneš et al., 2008). Their statistical foundations remain underdeveloped, however. The dominant practice is calibration; when estimation is attempted it is typically partial, with a subset of parameters sampled by random-walk Metropolis–Hastings and the rest held fixed. This gap reflects a computational obstacle. The QZ-based rational expectations solution (Klein, 2000) is discontinuous in model parameters, which rules out gradient-based samplers and confines practitioners to random-walk proposals that scale poorly with dimension.

This paper develops a Bayesian framework that addresses two obstacles to full inference. The first is computational. We replace QZ with the structure-preserving doubling algorithm (SDA; Huber et al., 2026), which yields a rational expectations solution that is smooth in the structural parameters. Exact gradients are obtained by automatic differentiation in JAX (Bradbury et al., 2018), allowing the No-U-Turn Sampler (NUTS; Hoffman and Gelman, 2014) to explore the joint posterior over structural coefficients and shock variances. The approach extends recent Hamiltonian Monte Carlo applications to DSGE estimation (Farkas and Tatar, 2020; Childers et al., 2022) to the semi-structural QPM class.

The second obstacle is practical. An efficient sampler cannot overcome poorly specified priors. In trend–cycle models the likelihood is shaped by the relative variances of cyclical and trend shocks, quantities for which practitioners have little direct intuition. Our Brazil QPM has 25 shocks, so specifying Inverse-Gamma priors for 25 standard deviations is a high-dimensional elicitation problem; done poorly, it produces multimodal posteriors and misidentified decompositions. We replace direct elicitation with a two-stage calibration pipeline. First, signal-to-noise ratios (SNRs), corrected for the AR(1) persistence of the cyclical and drift components following Harvey (1989), replace 50 Inverse-Gamma hyperparameters with one interpretable smoothness ratio per trend. Second, a fixed-point disturbance smoother (Koopman, 1993), applied only to cycle shocks with trend shocks pinned at their SNR modes, refines the cycle-shock standard deviations until they are consistent with the data-implied conditional moments. The iteration is a damped Expectation–Maximization algorithm (Dempster et al., 1977; Shumway and Stoffer, 1982): the E-step is the Koopman smoother and the M-step is a multiplicative log-step update. The converged values serve as empirical Bayes prior modes.

The contribution is methodological: we develop and validate the full framework, the

SDA + AD + NUTS sampler and the SNR plus Koopman calibration pipeline. The sampler makes full posterior inference over QPMs feasible; the calibration pipeline makes it practical by reducing a 50-dimensional hyperparameter problem to a small set of economically interpretable smoothness ratios that are then refined with the data. Because the framework applies to any semi-structural model in linear rational expectations form, the same workflow transfers across countries and specifications. The estimation code and replication scripts are at https://github.com/agonzago/brazil_qpm_replication.

We illustrate the framework with an open-economy QPM for Brazil (2004Q1–2025Q2). We first write a rich template with the transmission channels suggested by the data: distributed-lag (Koyck) real-rate and exchange-rate pass-through, terms-of-trade effects, a time-varying neutral real rate, and a trend-growth channel. We then estimate ten specifications, including restrictions of the template and prior- or data-robustness variants, and adopt the most parsimonious specification as the headline baseline (Model C, introduced in Section 4). Section 5 shows what estimation buys: 90% posterior impulse-response bands for the monetary policy shock at the four-quarter horizon are five to twelve times tighter than the prior bands across the four transmission variables, and the posterior median output gap falls inside the Banco Central do Brasil's Low–High envelope in about 84% of the sample. Section 6 shows that the baseline delivers stable unconditional forecasts relative to a Bayesian VAR, especially for the policy rate; as discussed in that section, this is not a fully recursive real-time horse race.

2 Model Representation

Standard QPMs in the Forecasting and Policy Analysis System literature (Berg et al., 2006; Beneš et al., 2008) eliminate trends analytically before estimation. We instead write the stationary (gap) block and the non-stationary (trend) block as explicit systems linked by additive measurement equations, as advocated by Canova (2014) and used by Del Negro et al. (2019). The model is written at a quarterly frequency, and all rates are annualised.

Let $\mathbf{x}_t^{gap}$ denote the gap variables and $\mathbf{x}_t^{trend}$ the trend variables. The model is

$$\mathbf{x}_t^{gap} = f(\mathbf{x}_{t-1}^{gap}, E_t[\mathbf{x}_{t+1}^{gap}], \boldsymbol{\varepsilon}_t^{gap}; \boldsymbol{\theta}^{gap}) \tag{1}$$

$$\mathbf{x}_t^{trend} = g(\mathbf{x}_{t-1}^{trend}; \boldsymbol{\theta}^{trend}) + \boldsymbol{\varepsilon}_t^{trend} \tag{2}$$

$$\mathbf{y}_t^{obs} = \mathbf{H}^{gap}\,\mathbf{x}_t^{gap} + \mathbf{H}^{trend}\,\mathbf{x}_t^{trend}, \tag{3}$$

where $\mathbf{H}^{gap}$ and $\mathbf{H}^{trend}$ are selection matrices. The Brazil QPM is block-triangular: trends

evolve autonomously and do not feed back into gap dynamics, so the state-transition matrix decomposes into independent blocks and cyclical dynamics are invariant to the trend specification.

The explicit two-block form has three advantages. It modularises the specification, so cyclical and trend dynamics can be revised independently. It lets the data inform trend stationarity through estimated convergence speeds. And it remains valid when the trend block does feed back into the cycle: the state-space representation is then no longer block-diagonal, but the likelihood remains linear and Gaussian and the pipeline of Section 3 applies unchanged.[1] The Brazil equations are in Section 4; the complete system is in Appendix J.

3 The Estimation Algorithm

Full Bayesian inference for the model of Section 2 requires a posterior sampler, a model solution that meets the sampler's smoothness requirements, and a procedure for constructing priors on the shock-variance block. We start with the sampler and work backward.

Notation. The gap and trend states are $\mathbf{x}_t^{gap} \in \mathbb{R}^{n_g}$ and $\mathbf{x}_t^{trend} \in \mathbb{R}^{n_t}$, with parameter blocks $\boldsymbol{\theta}^{gap}$ and $\boldsymbol{\theta}^{trend}$ and shocks $\boldsymbol{\varepsilon}_t^{gap}$ and $\boldsymbol{\varepsilon}_t^{trend}$. Both covariance matrices are diagonal with entries $\{\sigma_j^2\}_{j=1}^{n_\varepsilon}$. The observables $\mathbf{y}_t \in \mathbb{R}^{n_y}$ are linked to the states by $\mathbf{H}^{gap}$ and $\mathbf{H}^{trend}$ in eq. (3). Let $\boldsymbol{\theta} = (\boldsymbol{\theta}^{gap}, \boldsymbol{\theta}^{trend}, \{\sigma_j\}_{j=1}^{n_\varepsilon})$ and let $\pi(\boldsymbol{\theta})$ be its prior. The likelihood $p(\mathbf{y}_{1:T} \mid \boldsymbol{\theta})$ is evaluated with the Kalman filter, and the posterior is $\pi(\boldsymbol{\theta} \mid \mathbf{y}_{1:T}) \propto p(\mathbf{y}_{1:T} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta})$.

3.1 Posterior sampling: NUTS

We sample the joint posterior with the No-U-Turn Sampler (Hoffman and Gelman, 2014), a self-tuning Hamiltonian Monte Carlo variant (Neal, 2011; Betancourt and Girolami, 2015). NUTS selects the integration length with a no-U-turn criterion and adapts the step size and mass matrix during warmup, removing the path-length tuning of vanilla HMC. Recent DSGE applications (Farkas and Tatar, 2020; Childers et al., 2022) confirm that gradient-based HMC scales well with the dimension of $\boldsymbol{\theta}$, where random-walk Metropolis and partial-conditioning

[1] The converse case, in which the cycle feeds back into the trend (for example, investment booms that raise the capital stock and trend growth), also leaves the likelihood linear and Gaussian, but not the two-step calibration, whose cyclical scale factors and trend priors presume autonomous trends; we do not pursue that case here.

schemes degrade. All structural parameters and shock standard deviations are sampled jointly without manual blocking.

NUTS requires a log-posterior that is differentiable in $\boldsymbol{\theta}$ end-to-end. This requirement drives the rest of the algorithm.

3.2 Differentiable model solution: SDA with automatic differentiation

The likelihood requires solving the linear rational expectations system at every posterior draw. The QZ decomposition of Klein (2000) is non-differentiable in $\boldsymbol{\theta}$: the generalized-eigenvalue ordering used to identify stable and unstable subspaces can flip when eigenvalues cross the unit circle, and the LAPACK reordering routine has no native JAX-differentiable implementation.

We replace QZ with the structure-preserving doubling algorithm (SDA; Huber et al., 2026; Kockerols, 2023), which solves the quadratic matrix equation for the policy function through a smooth fixed-point iteration. The doubling map preserves the symplectic structure that anchors the unique stable solution without an explicit eigenvalue sort, and the policy matrices are infinitely differentiable in $\boldsymbol{\theta}$ wherever the Blanchard–Kahn conditions hold. Huber et al. (2026) find that doubling is faster than QZ at comparable accuracy and remains stable through the parameter ranges encountered in posterior sampling. Exact gradients are obtained by automatic differentiation in JAX (Bradbury et al., 2018); the implicit-function-theorem identity that closes the fixed point yields a vector–Jacobian product whose adjoint Lyapunov equation is solved by the same doubling machinery. Appendix A provides the derivation.

3.3 Prior calibration for the trend–cycle decomposition

The shock-variance prior is the binding practical constraint: relative trend and cycle variances determine the permanent–transitory split of each observable. With 25 shocks in the Brazil QPM, the prior is a 50-dimensional object. We therefore replace direct elicitation with a two-stage pipeline.

Stage 1: AR(1)-corrected SNR. Cyclical shock modes are set from the empirical first-difference volatility of each observable, with an analytical correction for the persistence of the cyclical component (Harvey, 1989). The calibration therefore does not use HP-filtered

moments; the HP filter is used only to pre-filter the world oil and food commodity gaps before they enter as observables (Appendix I). A cycle map assigns an observable y_i to a cyclical shock c_j and a scale factor s_j. A trend map assigns each trend shock τ_k to a reference observable $y_{\text{ref}(k)}$, an SNR, and a reference type (level or growth). A reference need not be a distinct series: in Brazil, the level shock to potential output and the shock to potential growth both use log GDP, with different reference types.

Let Δy_i be the first difference of observable i, ρ the assumed quarterly persistence of the gap component (we use $\rho = 0.85$), and ρ_g the persistence of the drift or growth component (we use $\rho_g = 0.90$). These are calibration judgements about prior modes; they are distinct from the structural persistence coefficients estimated later. The scale factors and SNRs are reported in Appendix B.2, Table 1. For each cyclical shock,

$$\text{mode}_{c,j} \;=\; \sqrt{\tfrac{1}{2}(1+\rho)} \cdot \text{std}(\Delta y_i) \cdot s_j. \tag{4}$$

Because s_j multiplies a standard deviation, $s_j \in (0, 1]$ is the fraction of observable i's first-difference volatility attributed to the stationary cyclical innovation; equivalently, s_j^2 is the corresponding variance share. The values $s_j \in \{0.20, 0.25, 0.30, 0.50\}$ used here (Table 1) map to cycle variance shares of roughly 4% to 25%. The factor is set per series and does not partition variance across shocks.

Trend modes follow from one signal-to-noise ratio SNR_{τ_k} per trend, which encodes how smooth the trend should be relative to the cycle in its reference observable. The same gap persistence ρ enters the level-reference formula because both corrections come from the AR(1) gap component:

$$\text{mode}_{\tau,k} \;=\; \frac{\text{std}(\Delta y_{\text{ref}})}{\sqrt{2\,(1-\rho)\,\text{SNR}_{\tau_k}}} \qquad \text{(level reference)}. \tag{5}$$

A level reference applies when the trend shock moves the level or long-run component tied to the reference, such as potential output, the equilibrium real exchange rate, a real-rate trend, or an inflation trend. A growth reference applies when the shock moves the drift that accumulates into a level trend; the calibration then uses the second difference and the drift persistence ρ_g:

$$\text{mode}_{\tau,k} \;=\; \sqrt{\tfrac{1}{2}(1+\rho_g)} \cdot \frac{\text{std}(\Delta^2 y_{\text{ref}})}{\sqrt{\text{SNR}_{\tau_k}}} \qquad \text{(growth reference)}. \tag{6}$$

The persistence correction matters: at $\rho = 0.85$, omitting the $\sqrt{1-\rho^2}$ factor understates the cycle-shock standard deviation by about 3.3 and biases the trend anchor downward by the same factor. The modeller specifies the persistences and one SNR per trend; the rest is mechanical. Appendix B derives the correction from Harvey (1989).

Stage 2: Koopman EM fixed point. The SNR step produces internally consistent priors but does not condition on the data. The second stage runs a damped Expectation–Maximization iteration on the cycle-shock variances, with the Koopman (1993) disturbance smoother in the E-step. In unit-variance shock space, the smoother returns, for each date t and shock j, the conditional mean $\hat{\varepsilon}_{t,j}$ and variance $V_{t,j} \equiv \mathrm{Var}(\varepsilon_{t,j} \mid y_{1:T}, \boldsymbol{\sigma}^{(k)})$ of the standardized shock at iteration k. The smoothed second moment $\hat{\varepsilon}^2_{t,j} + V_{t,j}$ equals one in expectation when the assumed σ_j is correct. The M-step is the multiplicative log-step

$$\log \sigma_j^{(k+1)} - \log \sigma_j^{(k)} \;=\; \tfrac{1}{2}\, d \,\cdot\, \log\big(\,\mathcal{A}_t[\hat{\varepsilon}^2_{t,j} + V_{t,j}]\big), \tag{7}$$

where $d \in (0, 1]$ is a damping factor and $\mathcal{A}_t$ aggregates per-period second moments over time for a fixed shock, producing a scalar realized second moment q_j. We use the time-series mean. Trend shocks remain pinned at their SNR modes; only cycle shocks update. The standard EM monotonicity theorem (Dempster et al., 1977; Shumway and Stoffer, 1982) covers the undamped step; the damped log-step does not inherit that guarantee in finite samples. Implementation details, including aggregator choices, clipping, bound freezing, deterministic detrending, and the diffuse-smoother requirement, are in Appendix B.

From calibrated standard deviations to priors. The two stages produce empirical-Bayes shock standard deviations $\{\sigma_j^*\}$, converted into Inverse-Gamma priors $\mathrm{IG}(\alpha_j, (\alpha_j + 1)\,\sigma_j^*)$ for NUTS. The prior mode is σ_j^*; the concentration α_j controls how tightly the prior is centred on it. We use $\alpha_j = 3$ for trend shocks, whose modes come from economic SNR choices and are not updated by the EM step, and $\alpha_j = 5$ for cycle shocks, whose variances have incorporated sample information. The pipeline therefore tightens only the cycle-shock priors and leaves the SNR-anchored trend priors deliberately more diffuse (Appendix B).

3.4 Likelihood evaluation

With smooth policy matrices and empirical-Bayes priors, the log-likelihood is evaluated by the exact diffuse Kalman filter and smoother of Durbin and Koopman (2012), the standard treatment for state-space models with unit-root states. The univariate observation step handles missing values natively, so the same routine accommodates ragged-edge releases and robustness samples without changing the state-space representation.

4 Application: An Open-Economy QPM for Brazil

We apply the framework to a semi-structural open-economy QPM for Brazil over 2004Q1–2025Q2. We first describe a rich template with the transmission channels suggested by the data, then estimate restrictions and variants and retain Model C as the headline baseline. Section 4.1 documents the motivating facts, Section 4.2 describes the structural blocks (the full system is in Appendix J), and Sections 4.3–4.5 present the data, specifications, and posterior results.

4.1 Brazilian Stylized Facts

Brazilian cyclical fluctuations display a stop-and-go pattern (Chauvet, 2002; Weller, 2018), driven largely by terms-of-trade shocks (Fernández et al., 2018; Schmitt-Grohé and Uribe, 2018) and global financial conditions transmitted through the risk premium and exchange rate (Catão et al., 2008). We extract cyclical components with the Hamilton (2018) regression filter, which avoids the end-of-sample bias of Hodrick–Prescott.

Fact 1: External factors drive the business cycle

Global commodity prices, measured by the Banco Central do Brasil's IC-Br index, co-move strongly with Brazil's GDP gap. The correlation is 0.71 in USD, which isolates the global shock, but falls to -0.09 in BRL (Appendix Figure 25); a flexible exchange rate absorbs much of the commodity shock in domestic-currency terms. The co-movement is strongest for energy commodities ($r = 0.78$ in USD), reflecting Brazil's transition to a net oil exporter. Global risk sentiment, proxied by the five-year CDS spread, coincides with the major downturns of 2008–09 and 2014–16 through capital flow reversals and tighter financial conditions (Bank for International Settlements, 2009; International Monetary Fund, 2012; Catão et al., 2008). These facts motivate the terms-of-trade channel in the IS curve and the UIP condition with an endogenous risk premium.

Fact 2: The risk premium and exchange rate transmit external shocks

External shocks reach the domestic economy mainly through the exchange rate, which responds to both risk premia and commodity prices. The Real[2] has a positive correlation

[2]Throughout, the exchange rate is defined as USDBRL, so an increase denotes a stronger dollar and a weaker Real.

($r = 0.20$) with the CDS spread and a negative correlation ($r = -0.11$) with USD commodity prices, consistent with the adjustment mechanism in Catão et al. (2008). These co-movements are most visible during the 2008–09 and 2014–16 crises (Appendix Figure 17). Exchange-rate movements pass through to domestic inflation through institutionally mediated channels. Administered energy prices delay the transmission of global oil shocks, while food prices are more market-determined and respond faster (Appendix Figure 18). This asymmetry motivates the Koyck filters ($\rho_{QZ,E}$ and $\rho_{QZ,F}$) and the disaggregated supply block (Minella et al., 2003; Camacho and Perez-Quiros, 2013).

Fact 3: Monetary policy is forward-looking under persistent inflationary pressures

The Selic rate and twelve-month-ahead Focus inflation expectations move together, consistent with a forward-looking Taylor rule of the type estimated by Minella et al. (2003). The co-movement is not a Fisher identity: the Selic moves more than one-for-one with the expected-inflation gap under a Taylor-principle rule. Despite frequent headline breaches driven by administered prices, food, and energy, the inflation-targeting framework has anchored expectations and supports a forward-looking Phillips curve with a slowly evolving inflation trend. The systematic counter-cyclical stance motivates a Taylor rule with smoothing (γ_1), a strong inflation response ($\gamma_2 > 1$), and output-gap stabilization (γ_4). Appendix Table 8 maps the three facts to these coefficients.

4.2 The Brazil QPM Template

The template is a semi-structural open-economy New Keynesian model with five blocks: aggregate demand; a disaggregated supply block with separate core, food, and energy Phillips curves; a forward-looking policy rule; an external block with a detailed risk premium; and a small U.S. block. Some distributed-lag features are switched off in the candidate specifications; Model C is the baseline. We report only the equations that carry distinctive features; Appendix J lists the complete system.

Aggregate demand. The output gap follows a hybrid IS curve with Koyck-filtered real-rate and terms-of-trade transmission:

$$\tilde{y}_t = \beta_1\, \tilde{y}_{t-1} + \beta_2\, E_t[\tilde{y}_{t+1}] - \beta_3\, q_{r,t} + \beta_4\, \tilde{z}_t + \beta_5\, \tilde{y}^*_t + \phi^{IS}_{tt}\, q_{tt,t} + n^{\tilde{y}}_t, \tag{8}$$

where $q_{r,t} = (1-\rho_{QRR})\tilde{r}_t + \rho_{QRR} q_{r,t-1}$ is a geometric Koyck filter of the real-rate gap (Koyck, 1954), $q_{tt,t}$ is the analogous filter of the terms-of-trade gap, $\tilde{z}_t$ is the real exchange-rate gap, $\tilde{y}^*_t$ is the U.S. output gap, and $n^{\tilde{y}}_t$ is a stationary demand shock. The filter separates on-impact responses from cumulative effects and gives a calibration-friendly split between short-run and long-run elasticities. The positive ToT coefficient ϕ^{IS}_{tt} reflects Brazil's commodity-exporter status (Fact 1).

Aggregate supply. Headline inflation is a composite of core, food, and energy. Core inflation follows a hybrid New Keynesian Phillips curve with expectations, inertia, demand pressure, and supply channels:

$$\begin{aligned}\pi^x_t = {} & \lambda_{1x}\, E_t[\pi^x_{t+1}] + (1-\lambda_{1x})\, \pi^x_{t-1} + \lambda_{2x}\, \tilde{y}_t + \kappa_z\, q_{z,t} \\ & + \lambda_{4x}\, \tilde{z}_t + \lambda_{5x}\, (\tilde{p}^{oil}_t + \tilde{z}_t) + \lambda_{6x}\, (\tilde{p}^{food}_t + \tilde{z}_t) + \phi^{PC}_{tt}\, tt_t + \varepsilon^{\pi^x}_t .\end{aligned} \tag{9}$$

Exchange-rate pass-through operates through a smoothed real depreciation $q_{z,t}$, a Koyck filter on $4(\tilde{z}_t - \tilde{z}_{t-1})$; commodity input costs enter in domestic-currency terms. Food and energy inflation evolve through their own autoregressive processes driven by core inflation and dedicated real-depreciation filters, and headline inflation is the weighted average of the three components (Appendix J).

Monetary policy. The Selic gap follows a forward-looking Taylor rule with smoothing:

$$i_t = \gamma_1\, i_{t-1} + (1-\gamma_1)\left(\pi_{t+1} + \gamma_2\, E_t[\pi^{x,4}_{t+3}] + \gamma_4\, \tilde{y}_t\right) + n^i_t, \tag{10}$$

where $\pi^{x,4}_t \equiv \frac{1}{4}\sum_{j=0}^{3} \pi^x_{t-j}$ is the year-on-year core-inflation gap, π_{t+1} is the one-quarter-ahead headline inflation gap, and n^i_t is a persistent policy shock. The Fisher equation $\tilde{r}_t = i_t - E_t[\pi_{t+1}]$ closes the real-rate block.

External sector. The real exchange rate gap satisfies a modified uncovered interest parity condition with partial adjustment (Beneš et al., 2008):

$$\tilde{z}_t = \tfrac{\rho_{ls}}{1+\rho_{ls}}\, \tilde{z}_{t-1} + \tfrac{1}{1+\rho_{ls}}\, E_t[\tilde{z}_{t+1}] - \tfrac{(1-\rho_{ls})\,\psi_{uip}}{4\,(1+\rho_{ls})}\left(\tilde{r}_t - \tilde{r}^*_t - \tilde{\rho}_t\right) + \varepsilon^{\tilde{z}}_t. \tag{11}$$

The real interest differential is better suited to economies with persistent inflation differentials than nominal UIP. The country risk premium has a persistent trend and a transitory gap $\tilde{\rho}_t$ that responds to the terms-of-trade gap (Fact 2).

Trend block. Level trends follow random walks with time-varying drift; rate trends mean-revert toward economically motivated steady states:

$$X_t = (1 - \alpha_X)\, X_{t-1} + \alpha_X\, X_{ss} + \varepsilon_t^X, \tag{12}$$

where $\alpha_X \in [0, 1]$ is the convergence speed; $\alpha_X = 0$ nests a random walk. The domestic equilibrium real rate $\bar{r}_t$ error-corrects toward a UIP attractor that combines the U.S. neutral rate, the equilibrium risk premium, and steady-state real depreciation.

Terms-of-trade trend and potential growth. For commodity exporters, persistent terms-of-trade movements affect potential output through investment, real wages, and tradable inputs (Fernández et al., 2018; Schmitt-Grohé and Uribe, 2018; Catão et al., 2008). Recent QPM implementations formalize this link (González et al., 2020; Demidenko et al., 2016). We let the terms-of-trade trend feed potential growth through its own growth state:

$$\bar{g}_t = (1 - \alpha_g)\, \bar{g}_{t-1} + \alpha_g\, g_{ss} + \kappa_{g,tt} \left(\bar{g}_{t-1}^{tt} - g_{ss}^{tt}\right) + \varepsilon_t^{\bar{g}}, \tag{13}$$

with

$$\bar{g}_t^{tt} = (1 - \alpha_{tt})\, \bar{g}_{t-1}^{tt} + \alpha_{tt}\, g_{ss}^{tt} + \varepsilon_t^{\bar{g}^{tt}}, \tag{14}$$

and $\overline{tt}_t = \overline{tt}_{t-1} + \bar{g}_{t-1}^{tt} + \varepsilon_t^{\overline{tt}}$. Setting $\kappa_{g,tt} = 0$ shuts off the coupling. In the headline treatment $\kappa_{g,tt}$ is estimated under a Gamma prior; Appendix C gives the prior and cross-specification sensitivity.

Measurement equations. The baseline uses 15 observables. Each is the sum of its gap and trend components (for example, $y_t^{obs} = \tilde{y}_t + \bar{y}_t$ for log GDP). The world oil and food price gaps and the U.S. output gap enter as pure gap variables, and the inflation target is observed directly as the inflation trend. The U.S. neutral real rate (Laubach and Williams, 2003; Holston et al., 2017) is observed as the foreign equilibrium real rate and anchors the domestic neutral rate.

4.3 Data and Transformations

The model is estimated on quarterly Brazilian data from 2004Q1 to 2025Q2. Domestic series include real GDP, headline IPCA and its core, food, and energy components, the Selic rate, and the nominal exchange rate, from IBGE and the BCB. Core inflation averages the five official BCB core measures (EX0, EX3, Trimmed Smoothed, Double Weight, and P55), which reduces dependence on any single exclusion or trimming rule. The country

risk premium is the five-year CDS spread, preferred to the EMBIG because it is a purer measure of sovereign default risk and responds faster to global financial conditions (Fontana and Scheicher, 2016; D'Agostino and Ehrmann, 2014). U.S. and commodity series come from FRED. Appendix I lists sources and transformations. The terms-of-trade observable enters the measurement system as the sum of the cyclical gap tt_t and the level trend $\overline{tt}_t$, $y^{obs}_{tt,t} = tt_t + \overline{tt}_t = 100 \log(\text{IC-Br USD})_t$.

4.4 Model Specifications

We estimate ten specifications (Appendix Table 13). Four are body specifications that vary the cyclical block; one masks the COVID-19 observations; one imposes a no-attractor restriction on the neutral rate; and four are prior- or data-robustness variants. The exercise is model selection, not a search for the richest specification.

Model A retains the full distributed-lag structure, including the real-rate Koyck filter ρ_{QRR}, the real-depreciation filter ρ_{QZ} on the core Phillips curve, and the energy- and food-specific filters, and estimates 74 parameters on 15 observables. Model B drops ρ_{QRR}, so the IS curve uses the contemporaneous real-rate gap. Model C, the baseline, also sets $\rho_{QZ} = \rho_{QZ,F} = \rho_{QZ,E} = 0$, the parsimonious structure used by the 4GM Colombia model (González et al., 2020) and the Demidenko et al. (2016) template. Model D replaces the year-ahead core inflation target with one-quarter-ahead headline inflation. The no-attractor restriction $\alpha_{rr} = 0$ is Model C0; the COVID mask and the four prior/data variants are described in Appendix H. All ten use the same NUTS configuration (4 chains, 1,000 warmup, 1,000 posterior draws), the same diffuse Kalman initialization, and the same SNR plus Koopman EM calibration.

4.5 Posterior Results

Appendix Table 2 reports prior modes, posterior means, and 68% highest density intervals for all parameters across the ten specifications. All specifications converge: the maximum split-$\hat{R}$ statistic (Vehtari et al., 2021) is 1.01, the minimum bulk effective sample size is 941, and the Blanchard–Kahn condition holds at every posterior draw.

Across body specifications, the posterior places the policy regime and aggregate-demand block away from the parameterizations commonly reported in the Brazil QPM literature. We read these as posterior estimates within the maintained QPM specification, not as formal structural-identification results. The Taylor rule is markedly more inertial and more

aggressive on inflation than the calibrated benchmarks (Minella et al., 2003; Correa and Minella, 2010), and the IS curve has less own-lag persistence. Minella et al. (2003) report forward-looking inflation coefficients of 2.05–2.32 with survey expectations and 2.71–5.70 with BCB forecasts; our posterior mean of 3.38[3] lies in the upper part of that range, and the implied impact response $(1 - \gamma_1)\gamma_2 \approx 0.28$ is conventional. These departures are qualitatively consistent across all ten specifications and move away from the prior modes, so they are driven by the data rather than the priors. Model D's alternative Taylor-rule timing leaves the policy coefficients qualitatively unchanged. The four robustness variants preserve the pattern; where prior concentration is most relaxed (RUB1), the inflation response γ_2 rises further, suggesting that the baseline estimate is a conservative lower bound on the data-implied response.

5 Model Properties and Policy Analysis

This section examines the dynamic and policy properties of the estimated model. Additional properties, including potential growth, the historical shock decomposition, exchange-rate pass-through, and cross-specification gap estimates, are in Appendix K. All quantities use 200 posterior draws (a systematic thinning of the 4,000 NUTS draws); shaded bands are 68% and 90% highest density intervals (HDI).

5.1 Impulse Responses

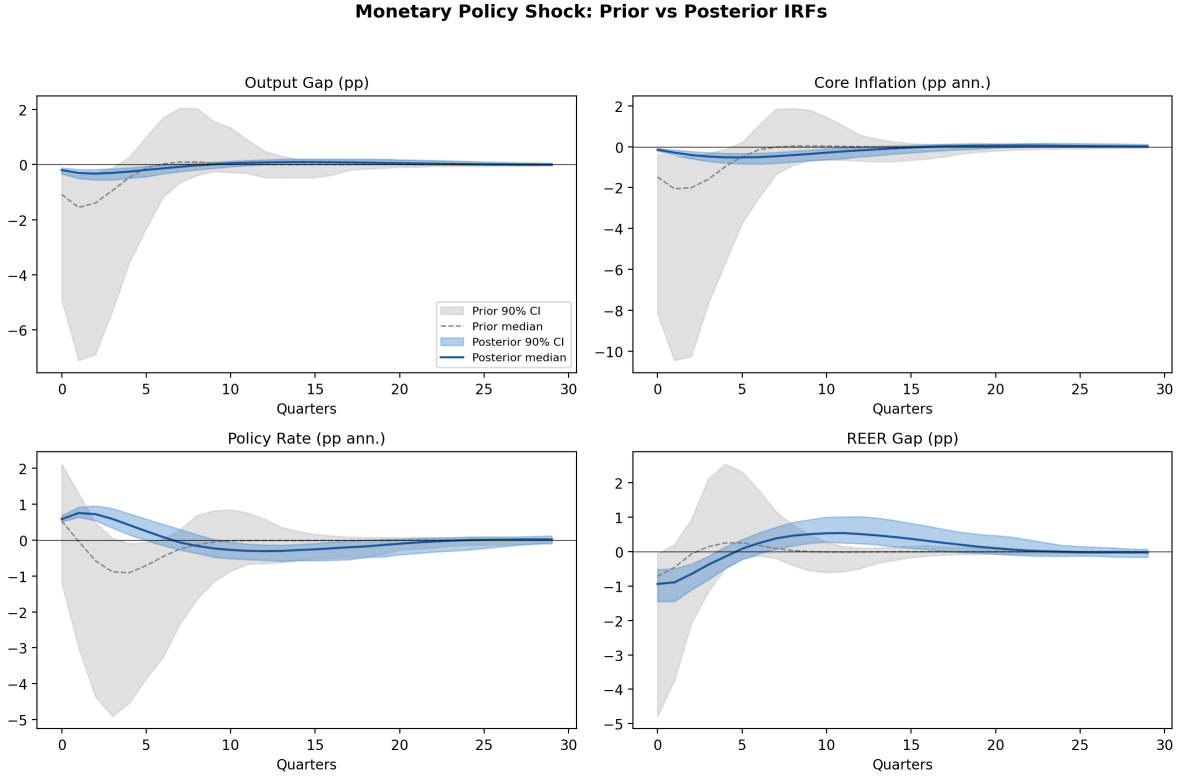
Figure 1 shows responses of the GDP gap, core inflation, the policy rate, and the real exchange rate gap to a monetary policy shock under Model C. A contractionary surprise opens a negative output gap, appreciates the Real, and produces core disinflation that peaks after roughly two years. The bands are tight: the 68% HDI of the impact policy-rate response is narrower than 0.15 percentage points despite joint inference over 71 parameters. The figure overlays the prior and posterior bands, so it also measures what estimation buys.

Estimation's payoff is visible in the same figure. The 90% prior band, sampled from the calibrated SNR pipeline before MCMC, is overlaid on the posterior band. At the four-quarter horizon, posterior bands are five to twelve times narrower than prior bands, with the largest compression for core inflation (twelvefold) and the output gap (tenfold). Under calibration alone the transmission responses carry substantial uncertainty; the data sharpen

[3]Parameters are summarised by posterior means; nonlinear functionals such as impulse responses and the sacrifice ratio by posterior medians, because their posteriors are skewed.

them to ranges useful for quantitative policy analysis. Responses to the remaining shocks are in Appendix F. This comparison is about transmission, not forecast accuracy; the forecast exercise in Section 6 compares the estimated QPM with a BVAR.

Figure 1: Monetary Policy Transmission: Prior vs. Posterior IRFs (Model C)

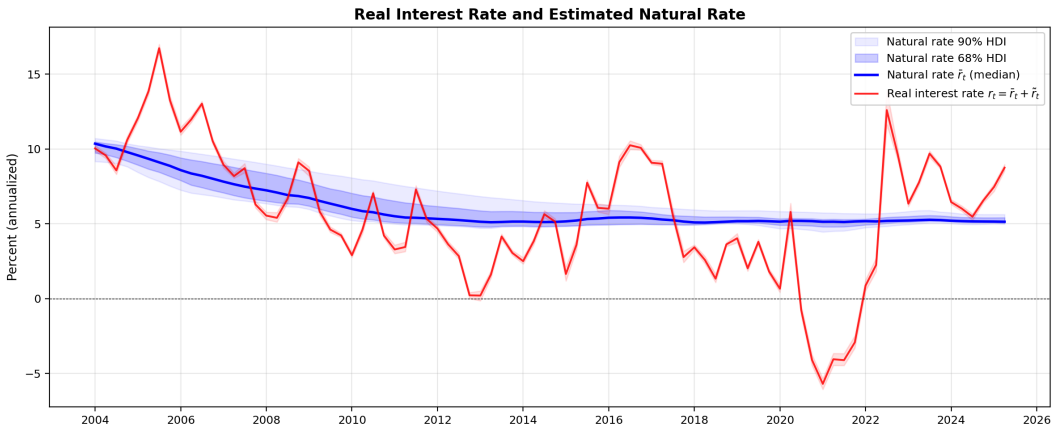


Note: 90% bands for Model C. Prior bands sample parameter vectors from the SNR-calibrated prior; posterior bands use 200 NUTS draws. Variables, left to right: GDP gap, core inflation gap, policy rate gap, real exchange rate gap.

5.2 Natural Real Rate

Figure 2 plots the estimated domestic equilibrium real rate $\bar{r}_t$ against the realized real rate. The posterior median falls from about ten percent in 2004–05 to roughly five percent by 2014, settles at 5.11% at the end of the sample, and has 68% HDI bands of about ± 0.6 percentage points. The decline matches the emerging-market evidence on falling neutral rates (Holston et al., 2017), while the level remains above the U.S. neutral rate because of Brazil's persistent country-risk premium. The end-of-sample value is in line with the Brazilian consensus documented by Banco Central do Brasil (2025). Across the time-varying-R^* body specifications, end-of-sample medians cluster within ± 0.04 percentage points.

Figure 2: Estimated Natural Real Rate (Model C)



Note: Posterior median (solid) and 68% HDI (shaded) for $\bar{r}_t$, with the realized real interest rate (dashed). Sample: 2004Q1–2025Q2.

To test the external anchor, we re-estimate Model C with $\alpha_{rr} = 0$. The restriction does not make $\bar{r}_t$ constant; it collapses the rate trend to a no-attractor random walk (Model C0). The data reject it: under the baseline, the 68% HDI of α_{rr} lies above zero, with a convergence half-life of about one year. Imposing $\alpha_{rr} = 0$ raises the end-of-sample median of $\bar{r}_t$ to 6.95%, 1.8 percentage points above baseline, and shifts the missing time-varying attractor onto the UIP semi-elasticity ψ_{uip}; the remaining coefficients are essentially unchanged.

The realized real rate overshoots the neutral rate during the disinflations of 2005–06, 2015–17, and 2023–25, consistent with the high smoothing and inflation response of the estimated Taylor rule (§ 4.5). Calibration would summarize these episodes with a point estimate; the posterior bands quantify the uncertainty around that judgement.

5.3 Taylor Rule Decomposition

Appendix Figure 23 decomposes the observed Selic rate into structural Taylor-rule components over 200 posterior draws. The identity

$$i_t^{obs} = \underbrace{\bar{r}_t + \bar{\pi}_t}_{\text{neutral rate}} + \underbrace{\gamma_1\, \tilde{i}_{t-1}}_{\text{smoothing}} + (1-\gamma_1)\Big(\underbrace{\pi_{t+1}}_{\text{expected infl.}} + \underbrace{\gamma_2\, \Delta^{A4} p^x_{t+3}}_{\text{infl. response}} + \underbrace{\gamma_4\, \tilde{y}_t}_{\text{output gap}} \Big) + \underbrace{\nu_t^i}_{\text{policy shock}} \tag{15}$$

holds at each date given the smoothed states. The right-hand terms are grouped into four series: a neutral-rate series combining $\bar{r}_t + \bar{\pi}_t$ with the smoothing term, an inflation series combining expected inflation with the year-ahead core response, an output-gap series, and a policy-shock residual.

The neutral-rate terms account for the level of the Selic and trace a secular decline from about 15% in the mid-2000s to roughly 8% by 2025. The inflation terms account for most cyclical variation, reflecting the posterior mean $\gamma_2 = 3.38$ (Appendix C). The output-gap contribution is modest, consistent with a small γ_4: the systematic response operates mainly through inflation. The decomposition also shows the operational role of the time-varying neutral rate; under the no-attractor restriction of § 5.2, the neutral-rate series would be smoother and the policy shock would absorb the structural variation captured by the attractor.

5.4 Inflation Expectations

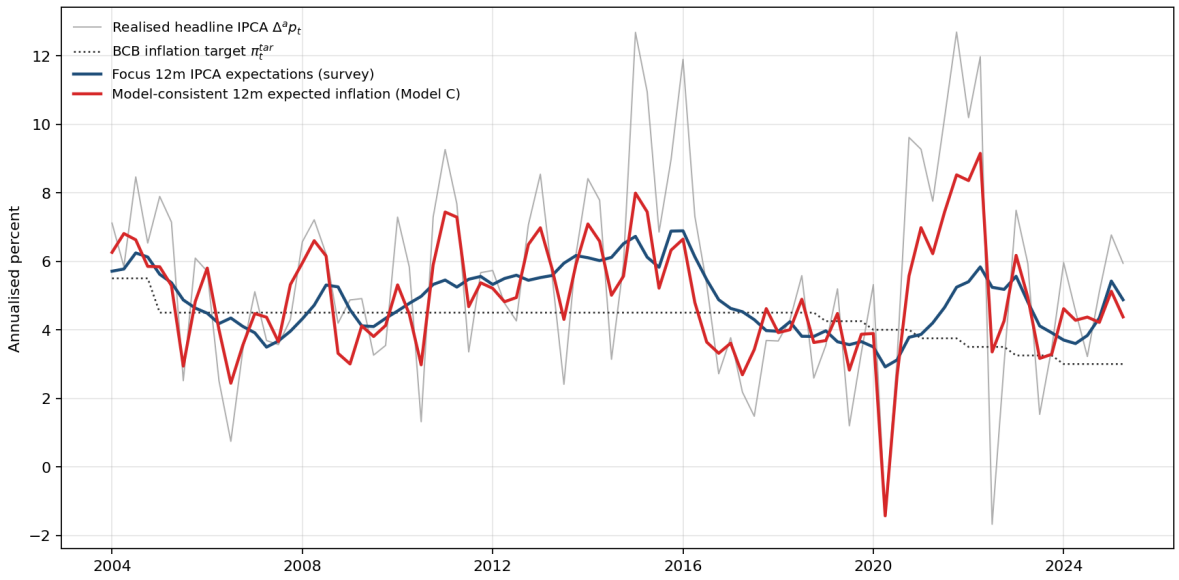
The smoothed inflation trend tracks the BCB target closely: it sits above target during the early disinflation of 2004–05, then matches each subsequent target step within 0.1 percentage points and ends at 3.04%, essentially the 3% target. Because the target enters as a direct measurement equation, the informative question is not whether the trend is identified but what expectations the policy rate responds to. The Taylor rule embeds next-quarter headline inflation and year-ahead core inflation, both evaluated at model-consistent forecasts rather than survey expectations. This matters for interpretation: the posterior $\gamma_2 \approx 3.5$ is a response to the model's own forecast, which need not coincide with the Focus survey.

Figure 3 plots the model-consistent 12-month-ahead headline expectation against the Focus series. The two series anchor near the BCB target and co-move at a contemporaneous correlation of +0.59. The Focus series is not used to identify Model C, so the co-movement is a validation of the model's forward-looking machinery rather than an outcome assumed by it. The Selic correlates more strongly with Focus (+0.46) than with the model-consistent 12-month expectation (+0.10). A reduced-form Taylor rule estimated against Focus would therefore assign a smaller inflation response; the structural γ_2 should be read against the model-consistent signal, which is a tighter forward-looking measure of inflation pressure than a smoothed survey average.

5.5 Output Gap and BCB Envelope Coverage

Figure 4 plots the posterior mean and median output gap against the Banco Central do Brasil's reference path, its low–high envelope, and its 25th–75th percentile band. The two posterior point estimates are essentially indistinguishable for Model C. Cross-specification gap paths are in Appendix Figure 20.

Figure 3: Model-Consistent vs. Survey Inflation Expectations (Model C)

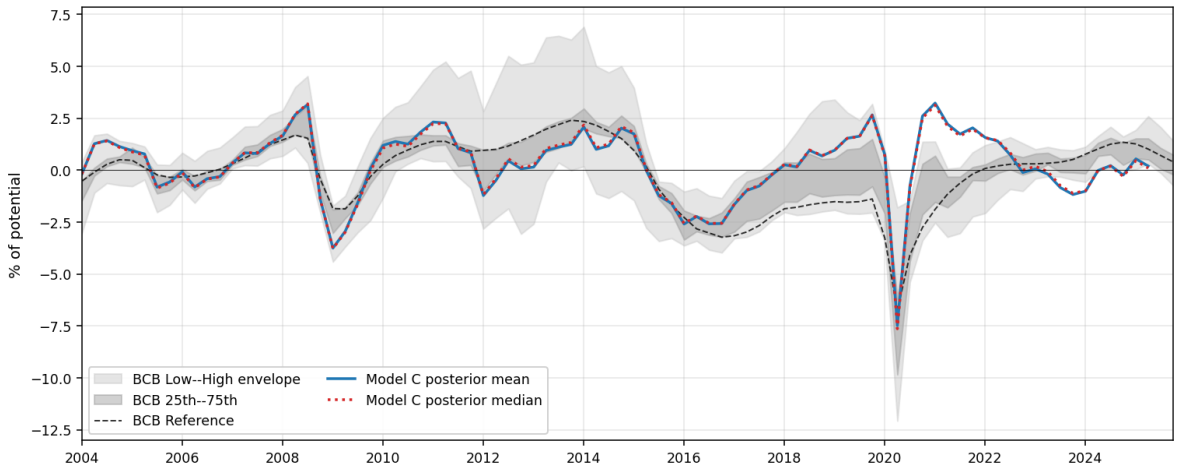


Note: Model-consistent 12-month-ahead headline inflation expectation (solid red), constructed by propagating the smoothed state of Model C forward through the rational-expectations transition matrix and averaging quarterly inflation over $t+1, \ldots, t+4$, against Focus survey 12-month-ahead IPCA expectations (solid blue) and the BCB target (dotted black). Realised headline IPCA (thin grey) is shown for reference; it is the annualised quarterly log-difference, $4\Delta \log p_t$, not a four-quarter difference. The model-consistent and survey series are 12-month (four-quarter) rates. All series annualised. Sample: 2004Q1–2025Q2.

The posterior traces a recognizable Brazilian cycle. The mid-2000s commodity boom opens a positive gap that peaks at +3.2 percentage points in 2008Q3, followed by a fall to −3.8 in 2009Q1. The 2014–16 recession produces a moderate negative gap that stays inside the BCB envelope. The 2020Q2 pandemic trough reaches −7.6 percentage points, well inside the BCB Low. The model diverges most from the envelope during the post-pandemic recovery (2020Q3–2022Q2), where the gap rebounds faster than the BCB's contemporaneous reading, and in 2023Q2–2024Q1, where the model's mildly negative gap sits below the BCB range.

The Model C posterior median falls inside the BCB low–high envelope in 83.7% of quarters, with a mean absolute deviation of 1.23 percentage points from the reference. Across specifications, coverage lies between 80.2% and 84.9% (Appendix Table 5). Where the gap leaves the band it does so by at most 1.5 percentage points in any quarter, and deviations during the major commodity-cycle episodes stay inside the BCB range; the 2020Q1–Q2 pandemic trough does too. The COVID mask brings the post-pandemic path closer to the

Figure 4: Posterior Output Gap (Model C) and BCB Envelope



Note: Posterior mean (solid blue) and median (dotted red) of the GDP gap for Model C, against the BCB reference (dashed), low–high envelope (light shaded), and 25th–75th band (darker shade).

envelope, suggesting that explicit treatment of 2020Q1–Q2 could narrow the residual; we leave that to future research.

5.6 Sacrifice Ratio

The sacrifice ratio measures the cumulative output cost per unit of cumulative excess inflation after a unit cost-push shock to the core-inflation gap, with the Taylor rule and the rest of the model responding endogenously over 120 quarters. At each posterior draw we shock the cost-push innovation, scale the response so the initial core-inflation gap rises by one percentage point, and compute

$$\mathrm{SR} \;=\; \frac{\left|\sum_{t=0}^{H} \tilde{y}_t\right|}{\sum_{t=0}^{H} |\pi_t^x|},$$

where $\tilde{y}_t$ and π_t^x are the output-gap and core-inflation-gap responses. The ratio is policy-conditional: it depends on the Phillips curve and on the Taylor-rule coefficients that determine the policy response (Berg et al., 2006). We report the within-model 68% HDI and the cross-specification region of posterior medians.

For Model C the posterior median is 0.37, with 68% HDI $[0.27, 0.51]$ and 90% HDI $[0.17, 0.60]$.[4] Across body specifications the posterior medians lie between 0.31 and 0.38,

[4]The reported figure uses the mixed convention, $|\sum_t \tilde{y}_t| / \sum_t |\pi_t^x|$. The symmetric convention, $|\sum_t \tilde{y}_t| / |\sum_t \pi_t^x|$, gives a median of 0.46 and the net convention 0.75. The v5 baseline implied 0.23; coupling the terms-of-trade trend raised the median to 0.37.

all inside Model C's 68% HDI, so cross-specification dispersion is small relative to posterior uncertainty.

A sign-restricted Bayesian VAR cross-check (Appendix G) gives a similar range: pre-COVID point estimates of 0.29 and 0.32 in the four-variable and five-variable specifications. Both reduced-form estimates lie inside Model C's 68% HDI and inside the cross-specification range. The agreement is reassuring, while the wide HDI shows that the data alone do not pin the disinflation cost sharply.

6 Out-of-Sample Forecasting Performance

We evaluate the estimated QPM through a pseudo-real-time filtering-and-forecasting exercise against a large Bayesian VAR (BVAR) benchmark available to a central-bank forecasting team (Litterman, 1986; Bańbura et al., 2010). Full recursive MCMC estimation of the QPM at every origin is infeasible at this dimension, so the exercise holds the full-sample posterior parameter vector fixed and recursively updates the latent state using only data available at each origin. The results are evidence on the forecast paths implied by the estimated specification, not a fully recursive real-time horse race. The full-sample posterior gives the QPM a modest informational advantage; a rolling-window QPM would face more parameter uncertainty, so the comparison is conservative for the paper's thesis.

The design is described in Section 6.1. Sections 6.2 and 6.3 report the unconditional and conditional results.

6.1 Exercise Design

The evaluation uses 24 quarterly origins from 2017Q1 to 2022Q4 and horizons $h = 1, 2, 3, 4, 8$. The targets are headline CPI inflation, core CPI inflation, GDP growth (all annualized quarter-on-quarter), and the Selic rate.

The BVAR includes 14 endogenous variables matching the QPM's observable set and a standard Minnesota prior with sum-of-coefficients and dummy-initial-observation shrinkage, with hyperparameters fixed at the Bańbura et al. (2010) defaults. Under Normal-inverse-Wishart conjugacy the posterior is closed form and is re-estimated at each origin on the expanding sample, with four lags and pandemic dummies. The conditional BVAR receives the realized future path of seven external and exogenous variables through the hard-constraint algorithm of Waggoner and Zha (1999) and forecasts the remaining domestic variables. The QPM forecasts use the full-sample posterior median parameters and re-run only the Kalman

filter at each origin, a common simplification in DSGE forecast evaluation (Del Negro and Schorfheide, 2013). The BVAR is therefore re-estimated recursively while the QPM uses full-sample parameters; we interpret the exercise as evidence on forecast stability and specification discipline rather than as a definitive ranking of operational systems. The QPM conditional forecast uses the same external conditioning set as the conditional BVAR.

6.2 Unconditional Forecast Performance

Appendix Table 9 reports unconditional RMSEs for the four QPM body specifications and the BVAR on the full sample; all forecast tables are collected in Appendix K.5. Table 10 repeats the exercise excluding the COVID-19 window. On the full sample, every QPM specification has a lower RMSE than the BVAR in every reported variable-horizon cell. Headline-inflation and GDP-growth errors stay stable across horizons, while the BVAR drifts sharply at medium horizons as the 2020 collapse and rebound propagate. For the policy rate, Model A's one-quarter-ahead RMSE is 0.54 percentage points against 1.28.

Excluding the COVID-19 window cuts every RMSE substantially (Appendix Table 10). The QPM has lower RMSEs at every horizon for headline and core inflation, narrowly at $h = 1$ and by a wider margin at $h \geq 2$. GDP-growth errors return to 2.15–2.72 at every horizon, against 4.96–9.57 for the BVAR, a ratio of two to four. The QPM also keeps lower policy-rate RMSEs at every horizon, with Model A best in every column. Among QPM specifications, Models A, B, and C cluster within about 5% of each other; Model D's one-quarter-ahead Taylor target costs 5–15% extra RMSE at long horizons for inflation and the policy rate. The results support Model C as the baseline: it is close to the best QPM alternative with a more parsimonious transmission structure.

6.3 Conditional Forecast Performance

Appendix Table 11 reports conditional RMSEs on the full sample, with each model receiving the realized future path of seven external and exogenous variables and forecasting the remaining domestic variables. Table 12, in Appendix K, repeats the exercise without the COVID-19 window.

The BVAR improves substantially when supplied with realized external paths: on the full sample its GDP-growth RMSE falls by 60–79% at $h \geq 2$ and its headline-inflation RMSE by roughly 56–83%. The QPM's analogous reductions are only 4–5% for GDP growth and 15–41% for headline inflation. The asymmetry indicates that the reduced-form benchmark

needs substantial ex post conditioning to produce reasonable paths in the stress sample, while the QPM's structural restrictions already impose much of that discipline unconditionally.

Stripping the COVID-19 window changes the picture. The BVAR's conditional gain on the COVID-excluded sample falls to 20–35% for inflation and 3–30% for GDP growth at $h \leq 4$, comparable to the QPM's inflation gain and bounded above by the QPM's slightly negative GDP-growth gain. The cross-class asymmetry in the full sample therefore reflects almost entirely the 2020 quarters. In tranquil samples the two approaches forecast at similar accuracy, with the QPM ahead on inflation and the policy rate; in the stress sample the BVAR's free forecasts are much more sensitive to the 2020 collapse and rebound. The QPM's expectations, disaggregated supply block, IS curve, and Taylor rule deliver more stable paths across regimes, so its performance depends less on realized future conditioning information.

In absolute conditional RMSE, the QPM outperforms the BVAR on headline inflation at $h = 1$ and $h = 2$; the BVAR catches up at $h = 3$–4 and pulls ahead at $h = 8$. For core inflation, Models B and C have the lowest RMSEs at every horizon, 10–20% below the BVAR. For GDP growth, conditioning reverses the unconditional ranking on the full sample: the QPM's conditional RMSE is about 5–10% higher than the BVAR's. Removing the COVID quarters restores the QPM's advantage at every horizon, widening at long horizons. The mechanism is that the QPM's structural IS curve anchors medium-run GDP growth, while the BVAR fits the unusual 2020–21 paths into its autoregressive coefficients and propagates them forward. For the policy rate, the QPM has lower RMSEs in both samples and at every horizon; the Taylor rule's structural identification of the systematic response is the source of the advantage.

Conditional rolling forecasts for Model C are shown in Appendix Figure 22.

7 Conclusion

This paper develops and estimates, via full Bayesian inference, a semi-structural Quarterly Projection Model for Brazil. The framework combines two ingredients. The first is the structure-preserving doubling algorithm with NUTS and automatic differentiation, which makes full posterior inference over all model parameters feasible and reliable. The second is the signal-to-noise calibration pipeline, corrected for AR(1) persistence following Harvey (1989) and completed by a Koopman disturbance-smoother fixed point (Koopman, 1993), which reduces the shock-variance prior problem to a small set of interpretable smoothness

ratios.

The Brazilian application (2004Q1–2025Q2, ten specifications estimated jointly) delivers structural and operational payoffs. On the structural side, and within the maintained QPM specification, the posterior implies a policy regime that is more inertial and more aggressive on inflation than the calibrated benchmarks, and an IS curve with less own-lag persistence; the pattern holds across the prior-sensitivity variants. The baseline sacrifice ratio is 0.37, with a 68% HDI of $[0.27, 0.53]$ that brackets the sign-restricted BVAR cross-check. On the operational side, the posterior median output gap falls inside the Banco Central do Brasil's low–high envelope in 83.7% of quarters. In the pseudo-real-time exercise the QPM produces stable unconditional paths, leads the recursively estimated BVAR at every horizon for the policy rate, and is more stable in the pandemic-stress sample; the reduced-form benchmark needs realized external conditioning to close much of the gap.

The framework ports: the open-source `qpm_toolbox` package specifies models in declarative `.qpm` files and exposes a high-level API, and the calibration pipeline is model-agnostic. The Brazil QPM (39 states, 71 parameters in the baseline Model C) runs four NUTS chains of 1,000 warmup and 1,000 posterior draws on a multi-core workstation using CPU-only JAX; the full set of results is reproducible in about two hours.

The 2020 quarters are the main unresolved measurement problem. A formal COVID-19 treatment through stochastic volatility, time-varying shock variances, or a regime-switching observation block would sharpen the 2020 estimates, although the COVID-mask robustness in Appendix H shows the headline results are stable. A fiscal block would allow analysis of fiscal–monetary interactions, which matter for Brazil. A more granular commodity block would capture the heterogeneous channels of terms-of-trade shocks (Schmitt-Grohé and Uribe, 2018; Fernández et al., 2018). More broadly, the framework applies to any semi-structural model in linear rational expectations form, so systematic Bayesian estimation across countries and specifications is now feasible.

A Numerical Methods

This appendix describes the computational methods that underpin the Bayesian estimation framework of Section 3. Three interrelated challenges are addressed: solving the rational expectations model at each proposed parameter vector, evaluating the log-likelihood via the Kalman filter, and computing the gradient of the log-posterior with respect to all estimated parameters, as required by the NUTS sampler. We first motivate the gradient requirement

and then describe each computational building block.

A.1 Why gradient-based sampling is necessary

Random-walk Metropolis–Hastings (RWMH) generates new parameter vectors by adding random noise to the current value: $\boldsymbol{\theta}^* = \boldsymbol{\theta}^{(s)} + \boldsymbol{\eta}$, $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. For a d-dimensional parameter space, the optimal proposal variance shrinks as $\sigma^2 \propto d^{-1}$ to maintain a reasonable acceptance rate (Roberts et al., 1997), producing highly autocorrelated chains that explore the posterior slowly. In our setting $d = 71$ for the baseline Model C, making standard RWMH impractical.

Hamiltonian Monte Carlo (HMC; Neal, 2011; Betancourt and Girolami, 2015) overcomes this dimensionality barrier by using the gradient of the log-posterior to guide proposals toward high-density regions. The method introduces auxiliary momentum variables $\boldsymbol{p} \in \mathbb{R}^d$ and defines a joint density proportional to $\exp\!\big(-U(\boldsymbol{\theta}) - \frac{1}{2}\boldsymbol{p}^\top \mathbf{M}^{-1} \boldsymbol{p}\big)$, where $U(\boldsymbol{\theta}) = -\log p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:T})$ is the negative log-posterior and $\mathbf{M}$ is a mass matrix. A proposal is generated by simulating L leapfrog steps with step size ϵ:

$$\boldsymbol{p}_{l+1/2} = \boldsymbol{p}_l - \tfrac{\epsilon}{2} \nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}_l), \tag{16}$$

$$\boldsymbol{\theta}_{l+1} = \boldsymbol{\theta}_l + \epsilon\, \mathbf{M}^{-1} \boldsymbol{p}_{l+1/2}, \tag{17}$$

$$\boldsymbol{p}_{l+1} = \boldsymbol{p}_{l+1/2} - \tfrac{\epsilon}{2} \nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}_{l+1}). \tag{18}$$

The gradient acts as a force that steers the proposal toward regions of high posterior density. NUTS (Hoffman and Gelman, 2014) automates the choice of trajectory length L, eliminating the main tuning parameter of HMC.

Equations (16)–(18) make the computational requirement explicit: every leapfrog step requires the gradient $\nabla_{\boldsymbol{\theta}} \log p(\boldsymbol{\theta} \mid \boldsymbol{y}_{1:T})$. For our model this gradient must propagate backward through the Kalman filter and the rational expectations solution, both of which depend on $\boldsymbol{\theta}$. Every building block of the pipeline must therefore be continuously differentiable in the structural parameters.

A.2 Model solution: the structure-preserving doubling algorithm

Writing the linearised model in canonical form

$$\mathbf{A}(\boldsymbol{\theta})\, E_t[\boldsymbol{x}_{t+1}] + \mathbf{B}(\boldsymbol{\theta})\, \boldsymbol{x}_t + \mathbf{C}(\boldsymbol{\theta})\, \boldsymbol{x}_{t-1} = \mathbf{0}, \tag{19}$$

where $\boldsymbol{x}_t$ collects all state variables, the stable rational-expectations solution is a matrix $\mathbf{T}$ satisfying the quadratic matrix equation

$$\mathbf{A}\,\mathbf{T}^2 + \mathbf{B}\,\mathbf{T} + \mathbf{C} = \mathbf{0}, \qquad \rho(\mathbf{T}) < 1. \tag{20}$$

Algorithm. The structure-preserving doubling algorithm of Huber et al. (2026) solves (20) via iterative matrix squaring. Initialise

$$\mathbf{E}_0 = -\mathbf{B}^{-1}\mathbf{C}, \qquad \mathbf{F}_0 = -\mathbf{B}^{-1}\mathbf{A}, \qquad \mathbf{X}_0 = \mathbf{E}_0, \qquad \mathbf{Y}_0 = \mathbf{F}_0. \tag{21}$$

At each iteration $k = 0, 1, \ldots$ the four matrices update simultaneously:

$$\mathbf{E}_{k+1} = \mathbf{E}_k\,(\mathbf{I} - \mathbf{Y}_k\,\mathbf{X}_k)^{-1}\,\mathbf{E}_k, \tag{22}$$

$$\mathbf{F}_{k+1} = \mathbf{F}_k\,(\mathbf{I} - \mathbf{X}_k\,\mathbf{Y}_k)^{-1}\,\mathbf{F}_k, \tag{23}$$

$$\mathbf{X}_{k+1} = \mathbf{X}_k + \mathbf{F}_k\,(\mathbf{I} - \mathbf{X}_k\,\mathbf{Y}_k)^{-1}\,\mathbf{X}_k\,\mathbf{E}_k, \tag{24}$$

$$\mathbf{Y}_{k+1} = \mathbf{Y}_k + \mathbf{E}_k\,(\mathbf{I} - \mathbf{Y}_k\,\mathbf{X}_k)^{-1}\,\mathbf{Y}_k\,\mathbf{F}_k. \tag{25}$$

The iterate $\mathbf{X}_k$ converges to the stable solution $\mathbf{T}$ at a quadratic rate: after k iterations the effective number of implicit fixed-point steps is 2^k, so convergence to machine precision is achieved in approximately $\lceil \log_2 n \rceil$ iterations (typically eight for our model). Termination is at $\|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F / \|\mathbf{X}_{k+1}\|_F < 10^{-12}$, and the solution is verified by checking the residual ratio of (20).

Advantages over QZ. Huber et al. (2026) benchmark SDA against the QZ decomposition of Klein (2000) on nearly 100 DSGE models from the Macroeconomic Model Data Base and show that SDA produces more accurate solutions in less computation time. As discussed in Section 3.2, SDA additionally offers smoothness: a continuous map of matrix multiplications and linear solves is differentiable in $\boldsymbol{\theta}$, while QZ involves an eigenvalue sort whose derivative is discontinuous where stable and unstable eigenvalues cross. An independent SDA implementation for DSGE solution appears in the Julia package `MacroModelling.jl` (Kockerols, 2023).

Solution uniqueness and the Blanchard–Kahn condition. The SDA converges to the minimal solvent of the quadratic matrix equation, the solution whose eigenvalues have smallest modulus. When the Blanchard–Kahn conditions hold (the number of eigenvalues inside the unit circle equals the number of predetermined variables) this minimal solvent is the unique stable rational-expectations solution (Klein, 2000). We compute the spectral radius

$\rho(\mathbf{T}_{\text{stat}})$ of the stationary block at every posterior draw; it remains below unity throughout the ten specifications, and any draw that would violate the stationarity condition receives a log-likelihood of $-\infty$ and is rejected automatically.

Differentiating the solution. The NUTS sampler requires the gradient of the log-posterior with respect to all structural parameters. We use implicit differentiation through the adjoint of the fixed-point condition. Differentiating $\mathbf{A}\mathbf{T}^{*2} + \mathbf{B}\mathbf{T}^{*} + \mathbf{C} = \mathbf{0}$ at the converged solution yields a generalised Sylvester equation

$$\mathbf{M}^{\top}\mathbf{W} + \mathbf{A}^{\top}\mathbf{W}\,\mathbf{T}^{*\top} = \mathbf{\Lambda}, \tag{26}$$

where $\mathbf{M} = 2\mathbf{A}\mathbf{T}^{*} + \mathbf{B}$, $\mathbf{\Lambda} = \partial\ell/\partial\text{vec}(\mathbf{T})$ is delivered by the Kalman backward pass, and $\mathbf{W}$ yields $\partial\ell/\partial\boldsymbol{\theta}$ via the chain rule. Equation (26) is solved by its own doubling iteration. The resulting vector-Jacobian product is registered as a `custom_vjp` in JAX (Bradbury et al., 2018), providing exact gradients to machine precision with constant memory cost.

A.3 Kalman filter initialisation: exact diffuse treatment

The state vector contains three exact unit roots (the log-levels of potential GDP, the real exchange rate trend, and the terms-of-trade trend), for which no proper unconditional distribution exists. The exact diffuse initialisation of Durbin and Koopman (2012), Chapter 5, decomposes the initial state covariance as

$$\mathbf{P}_{1|0} = \kappa\,\mathbf{P}_{\infty} + \mathbf{P}_{*}, \tag{27}$$

where $\kappa \to \infty$, $\mathbf{P}_{\infty}$ is a positive semi-definite matrix with rank equal to the number of diffuse states, and $\mathbf{P}_{*}$ captures the finite-variance component. The Kalman filter prediction-error decomposition is carried out analytically in both components, so the diffuse states are integrated out exactly without ever taking κ finite.

Following Koopman (1997), the multivariate filter is replaced by a sequence of univariate updates, one per observable per period. When the diffuse variance of an update is positive, the update reduces the rank of $\mathbf{P}_{\infty}$ by one; when it is zero, the standard Kalman update with $\mathbf{P}_{*}$ applies. The diffuse phase terminates at the first period d_{*} where $\mathbf{P}_{\infty} = \mathbf{0}$; in our application the diffuse phase terminates within the first few quarters of the sample. The policy-rate identity $RS_t = \bar{R}_t + \bar{\pi}_t + rs_t$ is encoded directly in the measurement equations, so the exact diffuse treatment operates on the same non-stationary subspace rather than initializing each state separately. The implementation is fully compatible with automatic differentiation in JAX, so the likelihood gradient is available end-to-end.

A.4 Steady-state filter via the DARE

The shock-variance calibration pipeline (Appendix B) runs the Koopman disturbance smoother at fixed parameter values. For that purpose we use a steady-state Kalman filter whose covariance is the solution of the discrete algebraic Riccati equation

$$\mathbf{V}^* = \mathbf{T}\,\mathbf{V}^*\,\mathbf{T}^\top + \mathbf{R}\,\mathbf{Q}\,\mathbf{R}^\top - \mathbf{T}\,\mathbf{V}^*\,\mathbf{Z}^\top\big(\mathbf{Z}\,\mathbf{V}^*\,\mathbf{Z}^\top + \mathbf{H}\big)^{-1}\mathbf{Z}\,\mathbf{V}^*\,\mathbf{T}^\top. \tag{28}$$

DARE is solved via a matrix doubling iteration analogous to the SDA. At the converged $\mathbf{V}^*$, the Fréchet derivative reduces to a discrete Lyapunov equation in the closed-loop transition matrix $\mathbf{A}_{\text{cl}} = (\mathbf{I}-\mathbf{K}\mathbf{Z})\mathbf{T}$, itself solved by doubling. The vector-Jacobian product is registered as a `custom_vjp` in JAX. In the posterior-sampling phase of the framework, the likelihood is evaluated via the exact diffuse filter of § A.3 rather than the steady-state filter.

B Shock Variance Calibration

This appendix details the two-stage calibration pipeline of Section 3.3 that produces empirical-Bayes priors for the structural-shock standard deviations.

B.1 Notation

The model contains n_c cyclical shocks $\varepsilon_t^{c_j}$ with standard deviations $\sigma_{c,j}$, and n_τ trend shocks $\varepsilon_t^{\tau_k}$ with standard deviations $\sigma_{\tau,k}$. Each trend shock τ_k is paired with a reference observable. The signal-to-noise ratio for trend shock τ_k is

$$\text{SNR}_{\tau_k} \;\equiv\; \frac{\sigma_{c,j}^2}{\sigma_{\tau,k}^2}, \tag{29}$$

so that SNR = 200 means the cyclical shock has 200 times the variance of its associated trend shock, enforcing a smooth trend.

B.2 Stage 1: AR(1)-corrected SNR

Let Δy_i denote the first difference of observable y_i, ρ the assumed cyclical persistence, and ρ_g the drift persistence. Both are calibration inputs, not estimated parameters: they are fixed judgements about how persistent the modeller believes the stationary gap (ρ) and the growth-rate or drift state (ρ_g) to be, and they enter only through the analytical AR(1) corrections that translate observed first-difference volatility into shock-innovation scales.

They are distinct from, and do not constrain, the structural persistence coefficients (such as β_1, ρ_{ls}, or ρ_{tt}) that the sampler estimates from the data. Two separate values are required because a gap is a stationary AR(1) component around a trend, whereas a growth reference is a drift state that accumulates into a level trend, so the two enter through the first and second difference of their reference series respectively. In the Brazil application (all body specifications) we set $\rho = 0.85$ for the macro gaps and $\rho_g = 0.90$ for the growth-rate states. For each cyclical shock j associated with observable y_i and scale factor s_j, the prior mode is

$$\text{mode}_{c,j} \;=\; \sqrt{\tfrac{1}{2}(1+\rho)} \cdot \text{std}(\Delta y_i) \cdot s_j, \tag{30}$$

and the corresponding Inverse-Gamma prior $\text{IG}(\alpha,\, (\alpha+1)\,\text{mode}_{c,j})$ uses $\alpha = 3$. Trend shock prior modes are derived from SNR_{τ_k}. For a level reference,

$$\text{mode}_{\tau,k} \;=\; \frac{\text{std}(\Delta y_{\text{ref}})}{\sqrt{2\,(1-\rho)\,\text{SNR}_{\tau_k}}}; \tag{31}$$

for a growth reference,

$$\text{mode}_{\tau,k} \;=\; \sqrt{\tfrac{1}{2}(1+\rho_g)} \cdot \frac{\text{std}(\Delta^2 y_{\text{ref}})}{\sqrt{\text{SNR}_{\tau_k}}}. \tag{32}$$

Calibration inputs for the baseline (Model C). Table 1 reports the two sets of numbers the modeller supplies for the baseline specification: the cyclical scale factors s_j of eq. (30) (Panel A) and the trend signal-to-noise ratios SNR_{τ_k} of eqs. (31)–(32) (Panel B), together with $\rho = 0.85$ and $\rho_g = 0.90$. The scale factors are deliberately below one. Because s_j multiplies a standard deviation in eq. (30), it is the fraction of observable i's first-difference standard deviation attributed to the stationary cyclical innovation, so s_j^2 is the corresponding first-difference variance share (for example $s_j = 0.30$ assigns about 9% of the first-difference variance to the cycle); the trend prior is set separately from a signal-to-noise ratio and is not constructed as the residual share. It is a share of first-difference volatility, not of the level variance, and it is set per series rather than partitioned across shocks. The trend SNRs span three orders of magnitude, from 30 for the risk-premium trend (a relatively volatile equilibrium) to 24,000 for the U.S. inflation trend (a near-constant anchor); a higher SNR enforces a smoother trend relative to its reference cycle. These are the only free inputs to Stage 1; everything downstream is mechanical.

The AR(1) correction. The factors $\sqrt{(1+\rho)/2}$ and $1/[2(1-\rho)]$ in (30)–(32) come from Harvey (1989), §2.5 and §6.1. For an AR(1) cyclical component $\tilde{x}_t = \rho\tilde{x}_{t-1} + \varepsilon_t$ with shock

Table 1: Stage 1 Calibration Inputs, Baseline Specification (Model C)

Panel A. Cyclical scale factors s_j

Observable	**Cyclical shock**	s_j
Real GDP (OBS_LGDP)	SHK_GDP_GAP	0.20
Headline CPI (OBS_DA_LCPI)	SHK_DA_LCPI	0.30
Core CPI (OBS_DA_LCPIX)	SHK_DA_LCPIX	0.30
Energy CPI (OBS_DA_LCPI_ENERGY)	SHK_DA_LCPI_ENERGY	0.50
U.S. CPI (OBS_DA_LCPI_US)	SHK_DA_LCPI_US	0.30
Exchange rate (OBS_DA_LS)	SHK_LZ_GAP	0.25
World oil gap (OBS_LRWOIL_GAP)	SHK_LRWOIL_GAP	0.50
World food gap (OBS_LRWFOOD_GAP)	SHK_LRWFOOD_GAP	0.50
U.S. output gap (OBS_GDP_GAP_US)	SHK_GAP_US	0.30
Risk premium (OBS_PREM)	SHK_PREM_GAP	0.50
Selic rate (OBS_RS)	SHK_RS	0.50
U.S. funds rate (OBS_RS_US)	SHK_RS_US	0.50
Terms of trade (OBS_TT)	SHK_TT_GAP	0.30

Panel B. Trend signal-to-noise ratios SNR_{τ_k}

Trend shock	**Reference / type**	SNR
SHK_LGDP_BAR	OBS_LGDP / level	50
SHK_G_TREND	OBS_LGDP / growth	200
SHK_LZ_BAR	OBS_DA_LS / level	200
SHK_DLZ_TREND	OBS_DA_LS / growth	400
SHK_PREM_BAR	OBS_PREM / level	30
SHK_PI_TREND	OBS_DA_LCPI / level	2 400
SHK_RR_BAR	OBS_RS / level	1 200
SHK_RR_BAR_US	OBS_RR_BAR_US / level	50
SHK_TT_BAR	OBS_TT / level	50
SHK_D_TT_BAR	OBS_TT / growth	50
SHK_PI_TREND_US	OBS_DA_LCPI_US / level	24 000

Note: Baseline specification (Model C) under the headline terms-of-trade-trend configuration. Panel A lists the scale factors s_j of eq. (30); Panel B the trend signal-to-noise ratios of eqs. (31)–(32), with the reference observable and reference type (level or growth). The remaining cyclical shock, food-inflation (SHK_DA_LCPI_FOOD), receives a directly specified Inverse-Gamma mode rather than a scale-factor entry, giving 25 structural-shock standard deviations in total. Persistences are $\rho = 0.85$ (gaps) and $\rho_g = 0.90$ (growth states).

variance σ_ε^2, the unconditional level variance is $\sigma_{\tilde{x}}^2 = \sigma_\varepsilon^2/(1-\rho^2)$ and, under the slow-trend approximation $\Delta y \approx \Delta\tilde{x}$,

$$\mathrm{Var}(\Delta y) \;\approx\; \mathrm{Var}(\Delta\tilde{x}) \;=\; \frac{2}{1+\rho}\,\sigma_{\tilde{x}}^2 \;=\; \frac{2}{(1+\rho)(1-\rho)}\,\sigma_\varepsilon^2. \tag{33}$$

Rearranging gives both the cycle-shock factor $\sigma_\varepsilon^2 = \frac{1}{2}(1+\rho)\,\mathrm{Var}(\Delta y)$ that appears in (30) and the level-variance factor $\sigma_{\tilde{x}}^2 = \mathrm{Var}(\Delta y)/[2(1-\rho)]$ that propagates into (31) through the SNR. The role of the persistence correction is therefore not optional: the unconditional cycle standard deviation is $\sigma_{\tilde{x}} = \sigma_\varepsilon/\sqrt{1-\rho^2}$, so identifying the cycle-shock standard deviation with $\mathrm{std}(\Delta y_i)$ directly conflates the shock-level volatility with the unconditional cycle-level volatility. At the realistic value $\rho = 0.85$ the correction factor $1/\sqrt{1-\rho^2} \approx 3.3$, so omitting it understates the cycle-shock standard deviation by that factor and, through the SNR step, biases the trend-shock anchor downward by the same factor.

B.3 Stage 2: Koopman disturbance-smoother fixed-point

Writing the state space in unit-variance shock space, the structural standard deviations are absorbed into the shock-loading matrix: $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{R} = \mathbf{R}_{\text{model}}\,\mathrm{diag}(\boldsymbol{\sigma})$. The Koopman (1993) disturbance smoother delivers smoothed estimates $\hat{\varepsilon}_{t,j}$ and posterior variances $V_{t,j}$. When the assumed σ_j is correct, the smoothed second moments $\hat{\varepsilon}_{t,j}^2 + V_{t,j}$ are dimensionless and equal one in expectation.

Per-iteration update. The EM map is the multiplicative log-step of eq. (7),

$$\log\sigma_j^{(k+1)} - \log\sigma_j^{(k)} \;=\; \tfrac{1}{2}\,d\,\cdot\,\log\big(\,\mathcal{A}_t[\hat{\varepsilon}_{t,j}^2 + V_{t,j}]\big),$$

where $d \in (0,1]$ is a damping factor (default 0.5) and $\mathcal{A}_t$ aggregates over t. The step is clipped to a maximum log-step $\log 3$ (preventing any single iteration from moving σ_j by more than a factor of three) and projected back into pre-specified σ bounds. The EM monotonicity theorem (Dempster et al., 1977; Shumway and Stoffer, 1982) guarantees marginal-likelihood improvement at each iteration for the undamped step; the damped log-step used here does not strictly inherit that guarantee in finite samples.

Trend pins (cycle-only EM). Trend shocks are pinned at their SNR-determined modes throughout the iteration; only the cyclical shocks update via (7). This matches the workflow at central banks running Dynare-based QPMs, where trend variances are anchored to economically motivated smoothness ratios and the data informs only the cycle variances.

Allowing the EM to update trend variances as well empirically produces large drifts in trend σ values because the EM has no economic anchor for them.

Aggregator $\mathcal{A}_t$. Three choices are available: the unweighted mean (the standard EM choice), the trimmed mean (drops the top and bottom α-fractions of periods before averaging), and the winsorised mean (clamps the extremes at the α-quantiles). All three coincide at convergence under correctly specified Gaussian shocks; the robust variants protect against anomalous periods (regime shifts, surprise crises) that would otherwise inflate the cycle-shock variance estimate. The Brazil application uses the unweighted mean; the robust variants are reported here only to document the toolbox.

Bound freezing. A shock j whose σ_j has hit a pre-specified bound for two consecutive iterations is frozen at that bound for the remainder of the EM. The two-iteration window prevents spurious freezing from a single transient update.

Deterministic detrending preprocessing. The transition matrix $\mathbf{P}_{\text{aug}}$ and intercept $\mathbf{c}_{\text{aug}}$ depend only on the structural parameters (held at their prior modes throughout the calibration), not on $\boldsymbol{\sigma}$. We compute the deterministic state path once before the EM loop, subtract it from the observation vector, and run the smoother on the deviation-form input. This keeps the steady-state Riccati recursion numerically well-behaved on unit-root systems.

Diffuse vs. standard smoother. For unit-root models, the diffuse smoother of Durbin and Koopman (2012) is required for unbiased trend-shock recovery: a non-diffuse prior on integrated trends causes initial-state mismatch to leak into early-period smoothed shocks, biasing trend-shock second moments upward and preventing convergence. The standard smoother is used only when a time-varying shock-scale schedule is required.

B.4 Inverse-Gamma prior construction

The two stages produce a vector of empirical-Bayes shock standard deviations $\{\sigma_j^*\}$, which are converted into Inverse-Gamma priors $\text{IG}(\alpha_j,\ (\alpha_j{+}1)\,\sigma_j^*)$. The concentration parameter α_j controls prior tightness: $\alpha_j = 3$ at the SNR stage produces weakly informative priors; $\alpha_j = 5$ after the EM step concentrates the prior more sharply. The pipeline therefore tightens priors only where the data have spoken: trend σ values stay at $\alpha = 3$ (SNR-determined modes), while cycle σ values move to $\alpha = 5$. The resulting priors enter the NumPyro NUTS model definition with no re-specification step.

Optional presample calibration. The calibration pipeline takes the data as input through the disturbance smoother. If a strict separation between prior construction and posterior inference is desired, the SNR and EM stages can be run on a presample of the data (a leading window, say the first half) and the resulting empirical-Bayes priors then carried into a posterior estimation on the full sample. In the application of Section 4 we run both stages on the full sample, consistent with standard empirical-Bayes practice.

Portability to Dynare. The calibration outputs the diagonal structural-shock covariance matrix $V_e = \mathrm{diag}(\sigma_j^{*2})$, which is distinct from the measurement-noise covariance H (set by the modeller from observation-error judgements). The forthcoming Dynare 8 (Adjemian et al., 2011) exposes V_e at the user-facing layer, so the SNR and Koopman EM pipeline can be reimplemented inside a Dynare workflow without modification to its internals.[5]

C Prior Distributions and Posterior Summaries

This appendix reports the prior distributions for the structural parameters of the Brazil QPM and the posterior means across the four body specifications (Models A, B, C, D) and the COVID-mask Model E. Cyclical and trend shock standard deviations receive Inverse-Gamma priors with modes set by the calibration pipeline of Appendix B; full prior-posterior distributions are available in the replication archive.

C.1 Prior families

Parameters bounded on $[0, 1]$ (persistence coefficients, smoothing weights, forward-looking shares, and trend convergence speeds) receive Beta priors with modes set to literature-informed calibrated values. Parameters that are non-negative and unbounded (Phillips curve slopes, Taylor rule coefficients, and pass-through elasticities) receive Gamma priors with modes matched to calibrated values. For two parameters whose calibrated mode is zero (λ_{4x} and ϕ_{tt}^{PC}) we use Half-Normal priors with mode at zero, treating these as restricted channels whose relevance is tested by the data. The 25 structural-shock standard deviations receive Inverse-Gamma priors with modes from the AR(1)-corrected SNR step (trend shocks, $\alpha = 3$) and the post-EM step (cycle shocks, $\alpha = 5$).

[5] Dynare 8 is forthcoming at the time of writing; the current stable release is Dynare 7.2.

Prior on the terms-of-trade trend coupling $\kappa_{g,tt}$. The loading of the trend-growth state on potential growth in eq. (13) is estimated under a Gamma prior with mode 0.020 and standard deviation 0.015, the Brazil-scaled counterpart of the Colombian calibration of 0.10 in González et al. (2020). The trend-growth steady state and convergence speed are calibrated at $g_{ss}^{tt} = 0$ and $\alpha_{tt} = 0.10$ respectively, both inactive over the sample average and matching the original 4GM (Colombia) parameterisation. The posterior mean of $\kappa_{g,tt}$ in the baseline specification (Model C) is 0.036 with standard deviation 0.013; across the ten specifications considered in the paper the posterior mean lies in $[0.031, 0.036]$, consistently shifted approximately one prior standard deviation above the prior mode. The sensitivity of the headline results to fixing $\kappa_{g,tt}$ at zero is reported in Appendix H.

C.2 Cross-specification posterior summary

Table 2 reports posterior means for a selected subset of structural parameters across the four body specifications (A, B, C, D) and Model E under the headline treatment in which $\kappa_{g,tt}$ is estimated. The pattern documented qualitatively in § 4.5 is visible here: the Taylor rule smoothing γ_1 clusters near 0.92, the long-run inflation response γ_2 above 3.0, and the IS-curve own-lag coefficient β_1 near 0.33, across all five specifications. The prior-sensitivity variants (RUB1, RUB3, RUB4, RUB5) and the no-attractor neutral-rate nested test (Model C0) are documented in Appendix H and § 5.2. The full spec-level summary (means, standard deviations, 68% HDIs, and convergence diagnostics) is in the replication archive.

The U.S. block's inflation coefficient is $\gamma_2^* = 0.86$. Because the U.S. rule already carries a unit inflation loading through the Fisher/neutral term, the total response to a persistent inflation gap is about 1.86, consistent with the Taylor principle.

C.3 Convergence diagnostics

All specifications converge cleanly under the Vehtari et al. (2021) split-$\hat{R}$ diagnostic: the maximum $\hat{R}$ across all parameters and all specifications is 1.01, the minimum bulk effective sample size is 941 (Model B), and the across-specification median bulk ESS exceeds 1,500. The Blanchard–Kahn stationarity condition is satisfied at every posterior draw across all specifications.

Table 2: Posterior Means, Cross-Specification (Selected Parameters)

	Posterior Mean by Model				
Parameter	A	B	**C**	D	E
IS curve					
β_1	0.340	0.320	0.325	0.333	0.382
β_3	0.145	0.144	0.139	0.145	0.083
ϕ_{tt}^{IS}	0.139	0.133	0.135	0.132	0.111
Phillips curve					
λ_{1x}	0.268	0.238	0.240	0.214	0.210
λ_{2x}	0.143	0.120	0.124	0.120	0.120
ϕ_{tt}^{PC}	0.026	0.026	0.027	0.025	0.021
Taylor rule					
γ_1	0.916	0.919	0.917	0.919	0.914
γ_2	3.490	3.370	3.380	3.130	3.080
γ_4	0.286	0.274	0.273	0.296	0.341
Exchange rate and risk premium					
ρ_{ls}	0.178	0.168	0.169	0.182	0.234
ψ_{uip}	1.050	0.930	0.960	0.760	0.529
$\rho_{\tilde{\rho}}$	0.888	0.870	0.878	0.850	0.810
Trend convergence speeds and TT-trend coupling					
α_π	0.002	0.002	0.002	0.002	0.002
$\alpha_{\Delta z}$	0.309	0.313	0.314	0.323	0.349
$\kappa_{g,tt}$	0.034	0.036	0.036	0.036	0.031

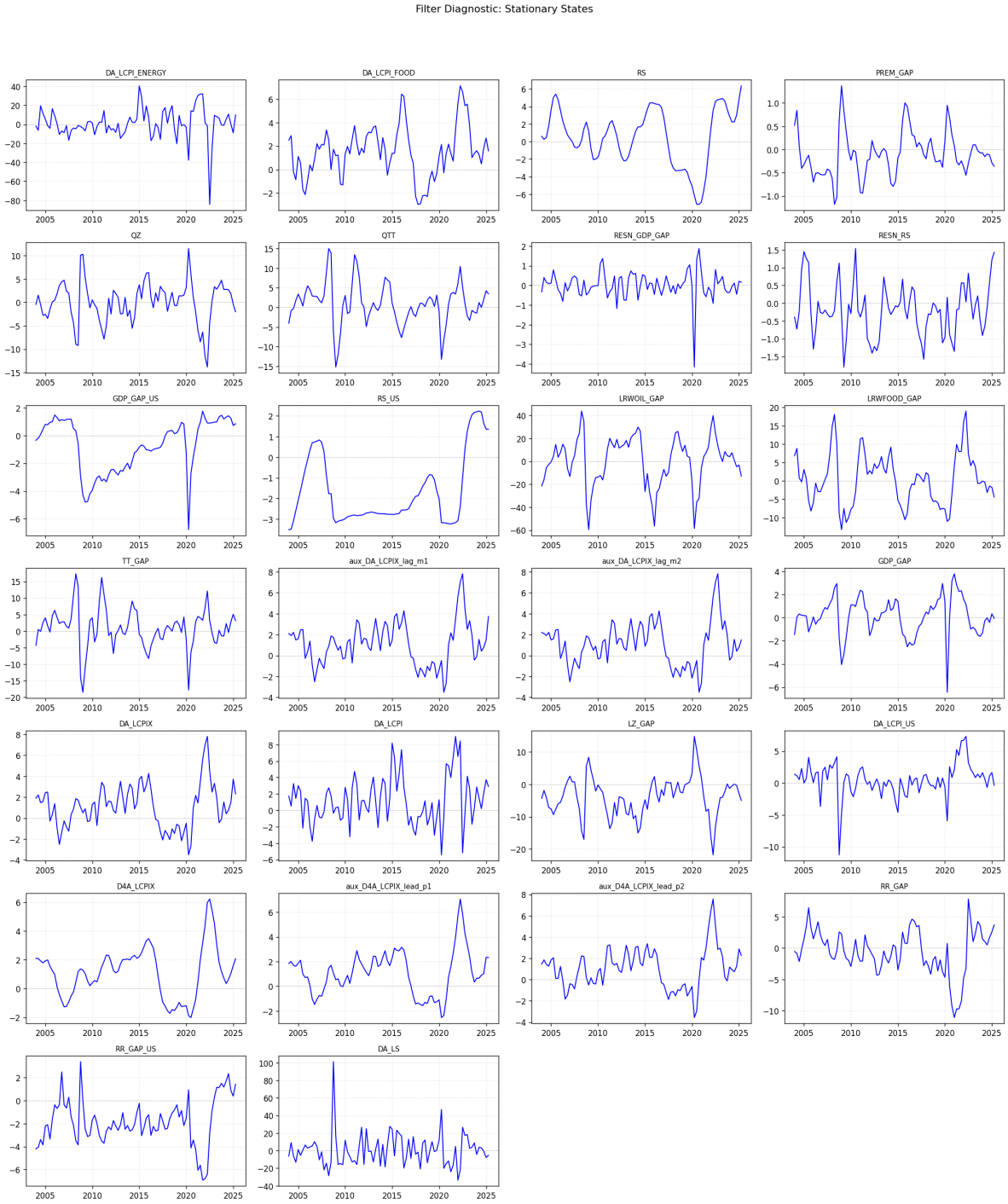
Note: Posterior means from 4,000 NUTS draws per specification (4 chains of 1,000 warmup and 1,000 samples each), under the headline specification with $\kappa_{g,tt}$ estimated. Baseline column (Model C) in bold. Convergence diagnostics: $\hat{R} \leq 1.01$ for every parameter in every specification; minimum bulk effective sample size is 941 (Model B). The no-attractor neutral-rate nested test on the baseline is reported in § 5.2. The full posterior summary (standard deviations and 68% HDIs) is available in the replication archive.

D Filtration and Estimation Results

This appendix reports the Kalman-smoothed estimates of the latent state variables under Model C: cyclical (gap) states, trend states, and the posterior-filtration overlay of the observed data against gap-plus-trend reconstructions.

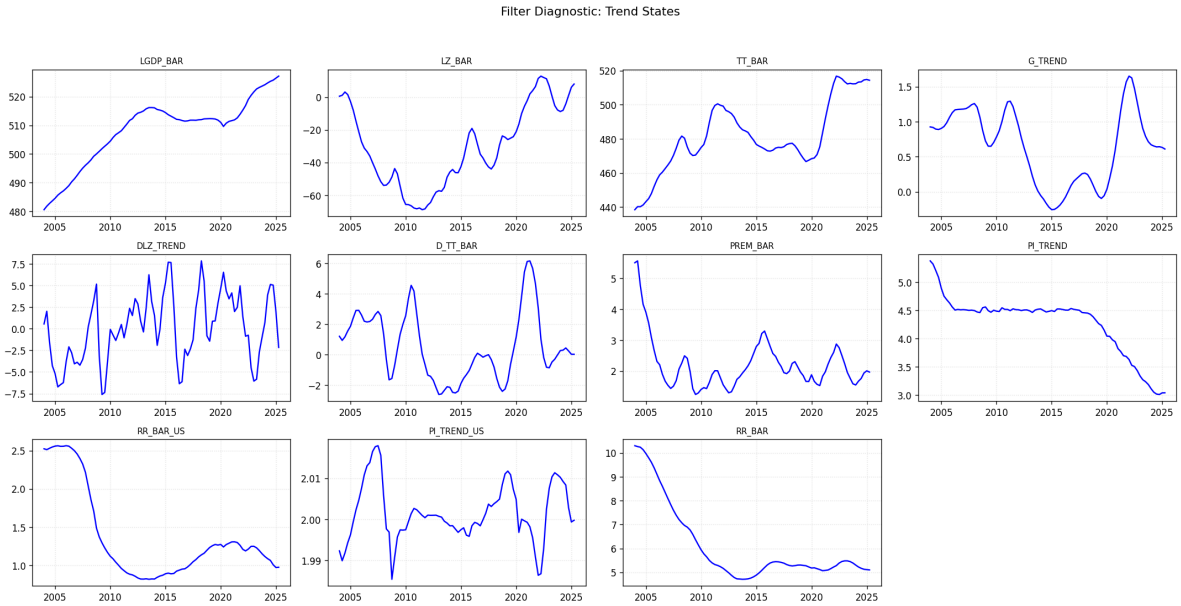
The smoothed states confirm the qualitative narrative of Section 4: a smoothly evolving trend block, including the gradually declining domestic equilibrium real rate of § 5.2, separated by the gap–trend block-triangular structure (§ 2) from a cyclical block that absorbs the high-frequency variation in the observables. Cross-specification smoothed states (not shown) are qualitatively similar across Models A through E; the per-specification figures and NPZ archives are in the replication package.

Figure 5: Kalman-Smoothed Cyclical States (Model C)



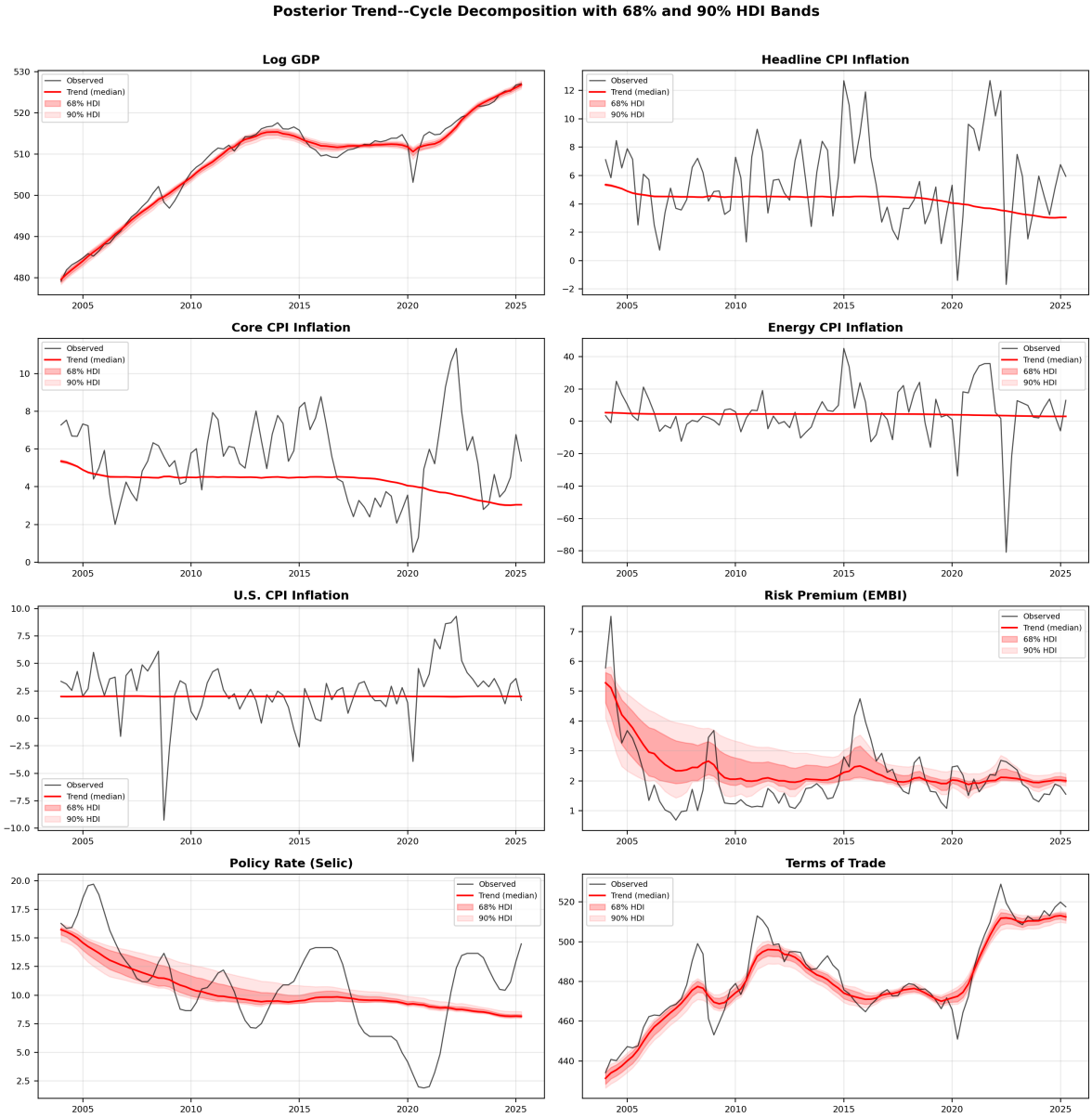
Note: Kalman-smoothed cyclical components evaluated at the posterior median. Sample: 2004Q1–2025Q2.

Figure 6: Kalman-Smoothed Trend States (Model C)



Note: Kalman-smoothed trend components evaluated at the posterior median. Sample: 2004Q1–2025Q2.

Figure 7: Posterior Filtration: Observed Series and Gap–Trend Reconstruction (Model C)



Note: For each observable, the observed series (solid) is plotted against the posterior-median sum of its gap and trend components. Shaded bands give the 68% HDI from 200 posterior draws.

E Model Property Definitions

This appendix defines the model-property and policy-experiment quantities reported in § 5.

Impulse response function (IRF). The response of variable y to a unit innovation in structural shock ε_j at horizon h is computed from the policy matrices $\mathbf{T}$ and the shock-loading $\mathbf{R}$:

$$\mathrm{IRF}_h(y, \varepsilon_j) \;=\; \mathbf{e}_y^\top \mathbf{T}^h \mathbf{R}\, \mathbf{e}_j,$$

where $\mathbf{e}_y$ and $\mathbf{e}_j$ are selection vectors. Bands shown in § 5.1 use 200 posterior draws.

Sacrifice ratio. Following the gap-form QPM convention of Berg et al. (2006) and Cecchetti and Rich (2001), the sacrifice ratio is the absolute cumulative output gap response divided by the summed absolute core-inflation gap responses to a unit cost-push shock ε^{π^x}, with the Taylor rule and the rest of the model responding endogenously:

$$\mathrm{SR} \;=\; \frac{\left|\sum_{h=0}^{H} \mathrm{IRF}_h(\tilde{y},\, \varepsilon^{\pi^x})\right|}{\sum_{h=0}^{H} \left|\mathrm{IRF}_h(\pi^x,\, \varepsilon^{\pi^x})\right|},$$

with $H = 120$ quarters. The IRF is rescaled so that the initial core-inflation gap rises by one percentage point. The quantity is policy-conditional: it inherits both the Phillips-curve parameters governing the inflation–gap trade-off and the Taylor-rule coefficients governing the central bank's response.

Highest density interval (HDI). For a posterior sample $\{x^{(s)}\}_{s=1}^{S}$, the α-level HDI is the shortest interval $[a, b]$ such that $\Pr(a \le x \le b) = \alpha$. We report 68% and 90% HDIs throughout.

BCB-envelope coverage. For a posterior sample of the output gap path $\{\tilde{y}_t^{(s)}\}_{s=1}^{S}$ and the BCB-published low and high envelope $[\tilde{y}_t^{\mathrm{low}},\, \tilde{y}_t^{\mathrm{high}}]$, the envelope coverage is the fraction of quarters in which the posterior median falls inside the envelope:

$$\mathrm{Coverage} \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\big\{\tilde{y}_t^{\mathrm{low}} \le \mathrm{median}_s\{\tilde{y}_t^{(s)}\} \le \tilde{y}_t^{\mathrm{high}}\big\}.$$

We report this statistic for the full sample and for the non-pandemic subsample (excluding 2020Q1–Q4).

F Additional Impulse Response Functions

Section 5.1 reports impulse responses of the baseline Model C to the monetary policy shock. This appendix reports the remaining structural shocks for Model C and the IRFs to the monetary policy shock for the alternative body specifications.

F.1 Additional shocks (Model C)

Figure 8: Model C: IRF to a Core Cost-Push Shock

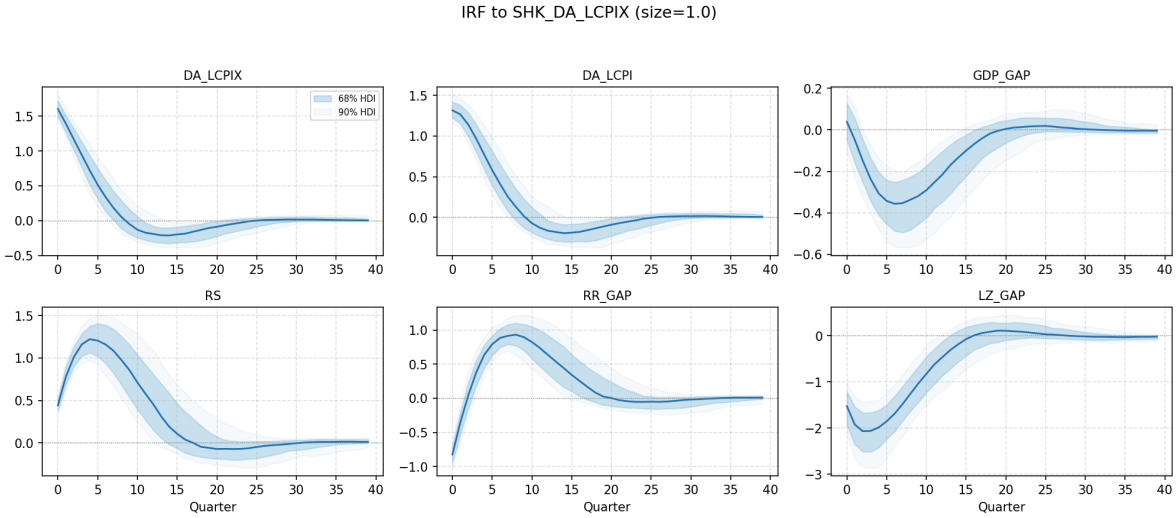


Figure 9: Model C: IRF to a Demand (Output Gap) Shock

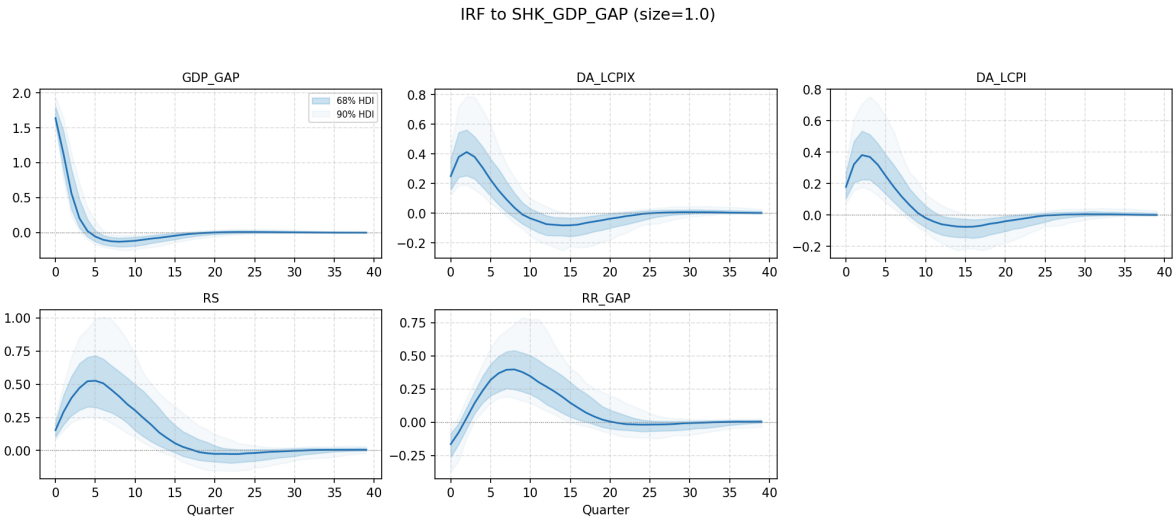


Figure 10: Model C: IRF to an Exchange-Rate Shock

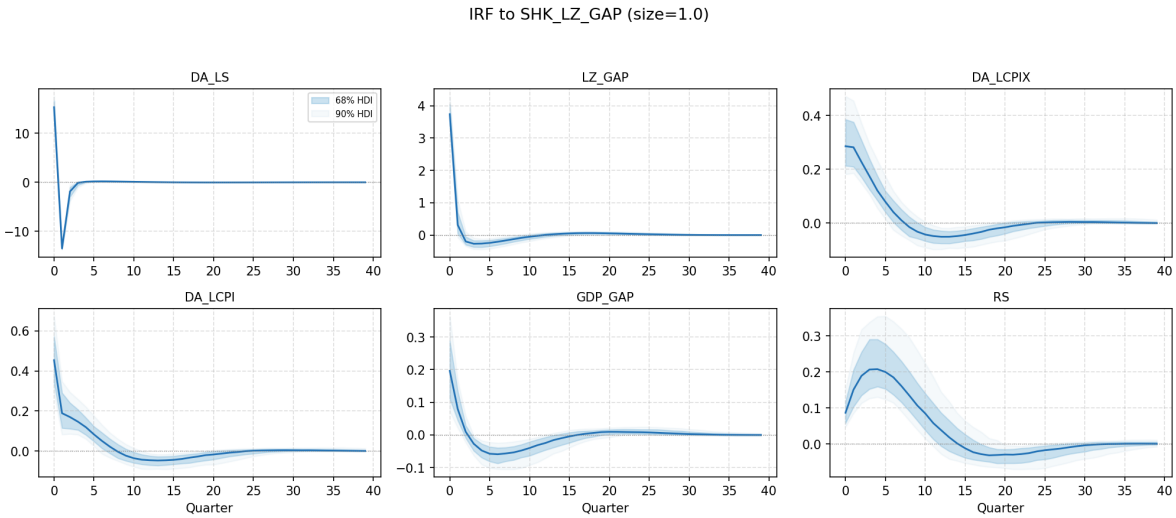
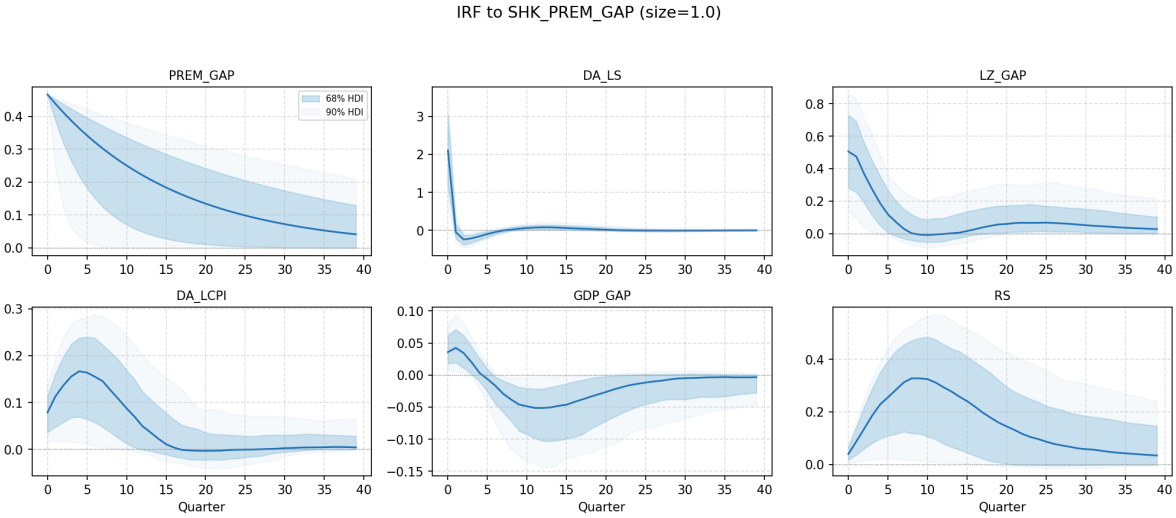


Figure 11: Model C: IRF to a Risk-Premium Shock



The terms-of-trade and risk-premium IRFs (Figures 12 and 11) confirm the empirical narrative of Fact 1 (Section 4.1): a positive ToT shock opens a positive output gap and appreciates the Real, while a positive risk-premium shock depreciates the Real, raises domestic real rates, and opens a negative output gap. The U.S. monetary policy shock (Figure 13) transmits through the UIP block, producing a depreciation and a contractionary domestic response.

Figure 12: Model C: IRF to a Terms-of-Trade Shock

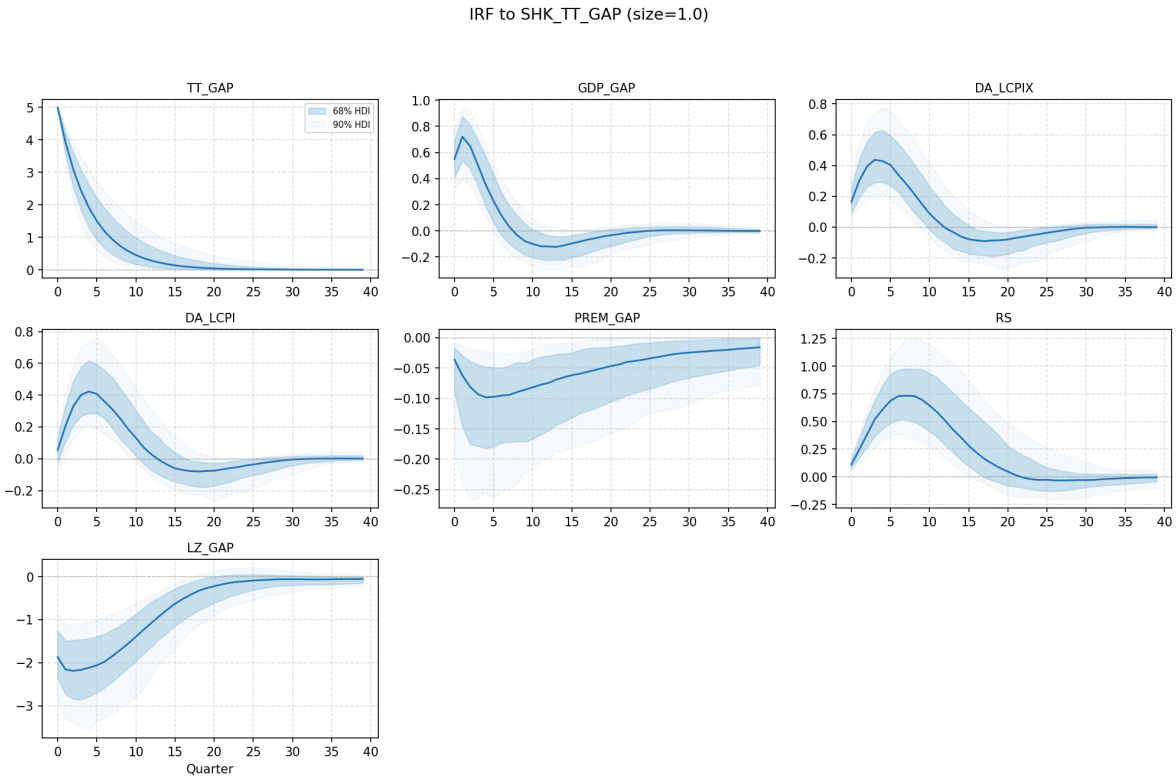
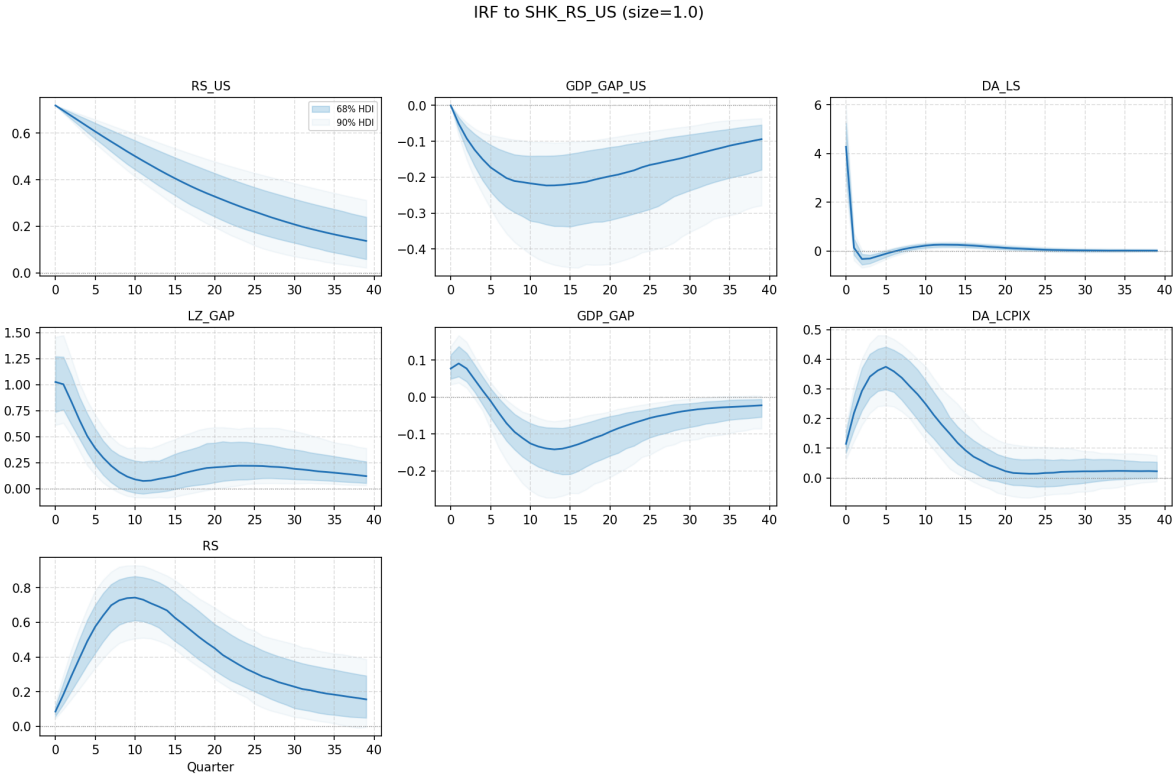


Figure 13: Model C: IRF to a U.S. Monetary Policy Shock



F.2 Cross-specification IRFs

The qualitative pattern of impulse responses is preserved across Models B, C, and D. The IRF figures for each alternative specification are stored in the replication archive at `model_<x>/irf/` where `<x>` runs over the body specifications; reproducing them for the reader of this paper is mechanical given the posterior draws.

G Reduced-Form Cross-Check: BVAR Sacrifice Ratio

As a reduced-form cross-check for the QPM sacrifice ratio (§ 5.6), we estimate a sign-restricted Bayesian vector autoregression (BVAR) on demeaned gap data for Brazil and compute the implied output cost of a unit reduction in inflation.

G.1 Specification

We estimate two BVAR variants. The *4-variable* BVAR includes the output gap, the headline inflation gap, the policy-rate gap, and the real exchange rate gap. The *5-variable* BVAR adds the core inflation gap. Both are estimated with two lags, a Minnesota prior (sum-of-coefficients and dummy-initial-observation shrinkage), and sign restrictions that identify a monetary policy shock by requiring a positive interest-rate response, a negative inflation response, and a negative output-gap response on impact, all maintained for one quarter. We report results on three samples: the full sample (2004Q1–2025Q2), the same sample after demeaning, and a pre-COVID window (2009Q1–2019Q4) that excludes the global financial crisis and the pandemic period.

G.2 Results

Table 3 reports the sacrifice ratio at horizon $h = 16$ (four years) for each specification and sample. The pre-COVID subsample produces sacrifice ratios of 0.29 (4-variable) and 0.32 (5-variable). Both lie inside the baseline Model C 68% HDI of [0.27, 0.51] (§ 5.6) and bracket the cross-specification region of posterior medians from below: Models A, B and C cluster at 0.36–0.38, Model D sits at 0.31, and Model E (COVID-mask) at 0.27. The full-sample BVAR estimates are smaller or negative, reflecting the contamination of reduced-form moments by the volatility of the 2020–2022 period that the structural QPM absorbs through its trend–cycle decomposition.

Table 3: BVAR Sacrifice Ratio (Sign-Restricted), $h = 16$

Sample	4-variable BVAR	5-variable BVAR
Full sample (2004Q1–2025Q2)	−0.08	0.04
Pre-COVID (2009Q1–2019Q4)	0.29	0.32

Note: Sacrifice ratio is the absolute ratio of the cumulative output-gap response to the cumulative inflation response at horizon $h = 16$, identified by sign restrictions on the monetary policy shock.

The pre-COVID BVAR estimates sit between Model E's posterior median (0.27, COVID-mask) and the body specifications' cluster (0.36–0.38), and lie comfortably inside the baseline Model C 68% HDI. The reduced-form cross-check is therefore consistent with the QPM finding of a moderate sacrifice ratio for Brazil, with the exact value depending on sample and specification.

The sign-restricted impulse responses for the monetary policy shock in both BVAR variants are reported in the replication archive at `bvar_sr_pt/figures/`.

H Sensitivity to Prior Specification and Sample

This appendix reports four prior-sensitivity and data-augmentation variants of the baseline (RUB1, RUB3, RUB4, RUB5), the COVID-masked Model E, and an additional tightened-prior check on ϕ_{tt}^{IS}. The four numbered variants are estimated on top of the Model C structural form (the baseline of § 4.4) under the same NUTS configuration as the body specifications (4 chains of 1,000 warmup and 1,000 posterior draws); all converge cleanly with $\hat{R} \leq 1.01$ and minimum bulk ESS above 2,300.

H.1 Robustness variants

Prior-sensitivity variants (RUB1, RUB3, RUB4). Table 4 reports three prior-sensitivity variants of the baseline. RUB1 doubles the dispersion of the priors on the four parameters with the largest prior-to-posterior movements (λ_{1x}, γ_2, α_{rr}, λ_{2x}). RUB3 weakens the Taylor-rule prior (mode shifted from 1.50 to 1.00, standard deviation widened). RUB4 fixes $\alpha_\pi = 0$, imposing a pure random-walk inflation trend. Across all three variants the posterior of γ_2 remains well above unity (indicating the Taylor-principle reading is data-driven rather than prior-driven), the inflation-trend convergence speed α_π stays near zero, and the BCB envelope coverage clusters at the baseline level (83.7%). Posterior γ_2 rises substantially under

RUB1 and RUB3, indicating that the baseline estimate is a conservative lower bound on the data-implied policy response.

Table 4: Prior-sensitivity variants of the baseline Model C

	Perturbation	γ_2	$\kappa_{g,tt}$	**BCB cov. (%)**
RUB1	Halved concentration on λ_{1x}, γ_2, α_{rr}, λ_{2x}	5.35	0.036	83.7
RUB3	Taylor prior: mode 1.00, sd 1.50	5.13	0.036	83.7
RUB4	$\alpha_\pi = 0$ (random-walk inflation trend)	3.32	0.036	83.7
Baseline (Model C)	—	3.38	0.036	83.7

Note: Posterior means under the headline specification with $\kappa_{g,tt}$ estimated. All three variants share the Model C structural form (drops ρ_{QRR}, $\rho_{QZ,E}$, $\rho_{QZ,F}$ Koyck filters). BCB envelope coverage is the share of quarters in 2004Q1–2025Q2 in which the posterior-mean output gap lies inside the BCB low–high band.

RUB5: Focus IPCA as additional trend observable. The Focus 12-month-ahead IPCA expectations series enters as a second measurement on the inflation trend ($\pi_t^{\text{Focus,obs}} = \bar{\pi}_t + \eta_t^{\text{Focus}}$, $\sigma_\eta^{\text{Focus}} = 0.6$ percentage points). Posterior results are virtually unchanged from the baseline (posterior $\gamma_2 = 3.47$, $\kappa_{g,tt} = 0.036$, BCB coverage 83.7%), demonstrating that the near-random-walk inflation trend is identified by data on actual inflation realisations and not driven by a single-observable identification strategy.

Tight ϕ_{tt}^{IS} prior. We re-estimate Model A with the Gamma prior on ϕ_{tt}^{IS} tightened from standard deviation 0.04 to 0.01 at the same mode of 0.05, concentrating the prior around its calibrated value. The posterior mean falls from 0.102 to 0.063 (HDI $[0.051, 0.072]$, $\hat{R} = 1.00$, $\text{ESS}_{\text{bulk}} = 5{,}946$), with non-overlapping 68% HDIs across the two specifications. The smoothed output gap shifts by at most 0.4 percentage points at any quarter over 2022Q4–2025Q2, and the sign of the gap is preserved at every quarter. The historical shock decomposition shows the ToT contribution and the residual IS shock $n_t^{\tilde{y}}$ shrinking in parallel, indicating that ϕ_{tt}^{IS} identifies the cancellation between these two channels rather than the level of the smoothed gap.

H.2 Model E: COVID-masked alternative

Model E is Model D with the COVID-19 observations masked (inflation, policy-rate and exchange-rate observations in 2020Q1–Q2 set to missing) so the Kalman filter handles them

via the diffuse-smoother NaN treatment of § A.3. Because the COVID quarters are masked, Model E is not included in the rolling forecast-evaluation exercise of § 6; the smoothed gap path and headline posterior numbers reported in this appendix show that masking the COVID quarters delivers a similar BCB-envelope coverage to the baseline (84.9 percent versus 83.7 percent) and similar end-of-sample $\bar{r}_t$ (5.22% versus 5.11%), confirming that the paper's headline results are robust to the choice not to treat COVID specially in the body.

H.3 Cross-specification BCB-envelope coverage

Table 5 reports the share of quarters in which the posterior median output gap lies inside the BCB-published low–high envelope, computed for each of the ten specifications.

Table 5: BCB-Envelope Coverage Across Specifications, $\kappa_{g,tt}$ estimated

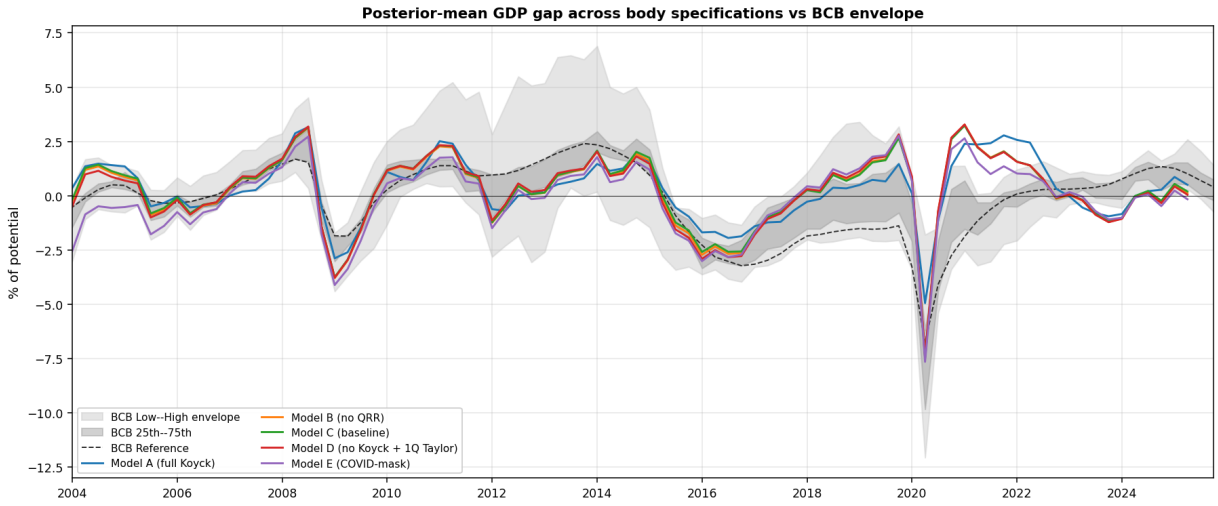
Specification	**All sample (%)**	**Non-pandemic (%)**
Model A	80.2	80.5
Model B	83.7	85.4
Model C (baseline)	**83.7**	**85.4**
Model D	83.7	85.4
Model E	84.9	86.6
RUB1	83.7	85.4
RUB3	83.7	85.4
RUB4	83.7	85.4
RUB5	83.7	85.4

Note: Share of quarters in which the posterior-mean output gap lies inside the BCB's low–high envelope. Sample 2004Q1–2025Q2 (86 quarters); non-pandemic sub-sample excludes 2020. All specifications are estimated under the TT-trend coupling specification with $\kappa_{g,tt}$ estimated under a Gamma prior centred at 0.020.

The coverage statistic is tightly clustered across specifications: all but Model A lie at 83.7% or 84.9% on the full sample and at 85.4% or 86.6% on the non-pandemic subsample. Model A (with the full distributed-lag IS-curve structure) falls outside this band at 80.2%, indicating a modest cost of retaining the QRR Koyck filter. Figure 14 overlays the posterior-mean GDP-gap path of each body specification on the BCB envelope; the five paths are tightly clustered for most of the sample, with the largest cross-specification dispersion concentrated in the 2008–09 financial-crisis trough and the 2020Q2 pandemic trough.

The result that the baseline Model C's gap reading is broadly consistent with the BCB's own assessment is therefore robust to the alternative specifications considered, including the four prior-sensitivity and data-augmentation variants estimated on top of the same Model C structural form. The no-attractor neutral-rate nested-restriction test, Model C0, is reported in § 5.2.

Figure 14: Posterior-mean GDP gap across body specifications vs BCB envelope



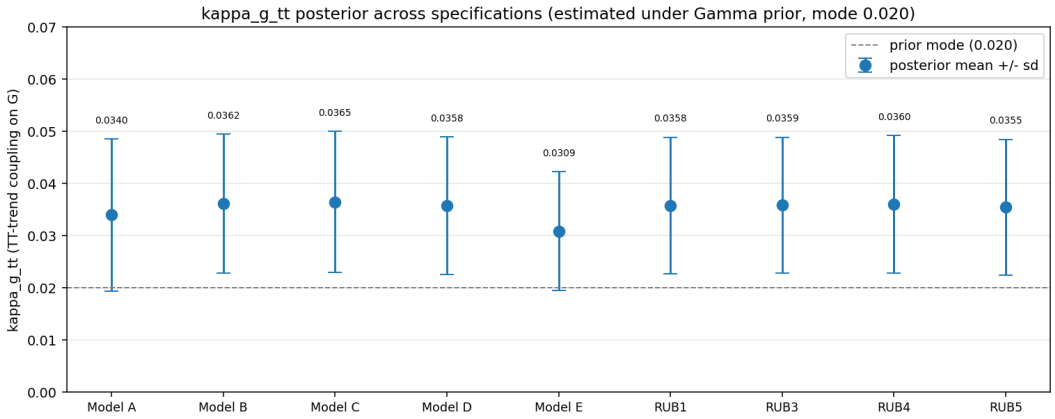
Note: Posterior-mean GDP gap for each of the four body specifications plus the COVID-mask Model E, plotted against the BCB's published reference (dashed) and low–high envelope (light shaded), with the 25th–75th band in darker shade. Sample: 2004Q1–2025Q2.

H.4 Sensitivity to the terms-of-trade trend coupling channel

This subsection reports the sensitivity of the headline results to the treatment of the loading coefficient $\kappa_{g,tt}$ in eq. (13). We compare two treatments: (i) $\kappa_{g,tt} = 0$, which switches off the trend-growth channel and decouples potential output growth from the terms-of-trade trend, and (ii) $\kappa_{g,tt}$ estimated under a Gamma prior centred at 0.020 with standard deviation 0.015, the headline treatment used throughout the paper.

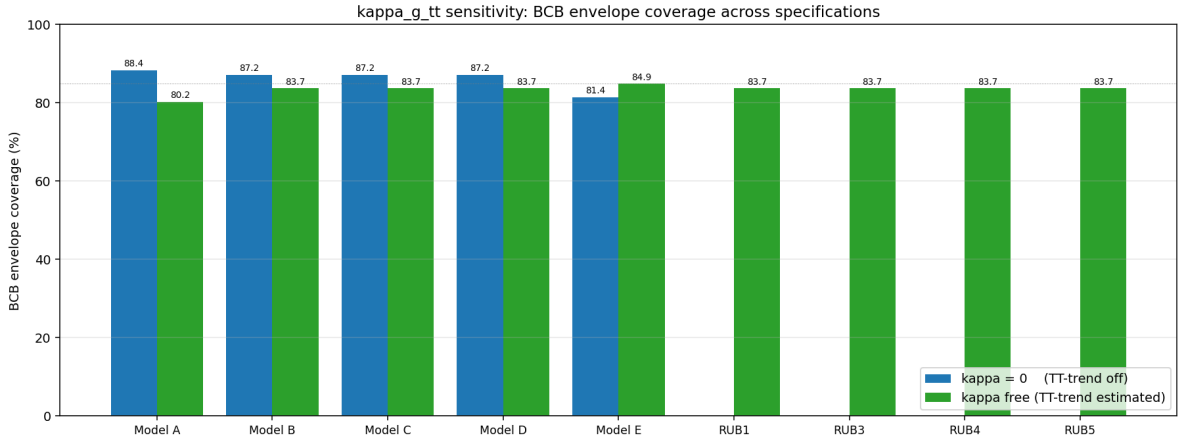
Across the specifications considered, both treatments deliver BCB envelope coverage within ±4 percentage points of one another; the posterior of $\kappa_{g,tt}$ under treatment (ii) settles at 0.035 ± 0.013 (mean of the per-specification posterior means and standard deviations) regardless of the IS-curve structure, supply-block formulation, prior treatment or COVID-masking choice (Figure 15). Figure 16 reports the BCB envelope coverage across the two treatments as a side-by-side bar chart; Figure 20 of the body reports the cross-specification posterior gap paths with the two treatments stacked side-by-side.

Figure 15: Posterior of $\kappa_{g,tt}$ Across Specifications



Note: Posterior mean of $\kappa_{g,tt}$ (point) with one-posterior-standard-deviation error bars, for each specification, under treatment (ii). Horizontal line: prior mode 0.020.

Figure 16: BCB Envelope Coverage, Two $\kappa_{g,tt}$ Treatments



Note: BCB envelope coverage (share of quarters inside the BCB low–high band) for each specification under two treatments of $\kappa_{g,tt}$: zero (TT-trend off) and estimated under Gamma prior.

The data is weakly but not negligibly informative about the magnitude of the coupling: the posterior of $\kappa_{g,tt}$ shifts consistently above the prior mode across specifications, by approximately one prior standard deviation, while the posterior standard deviation does not narrow relative to the prior. Together with the BCB coverage stability, this supports the conclusion that the headline results of the paper are robust to fixing $\kappa_{g,tt}$ at zero.

I Data Sources and Definitions

The model is estimated on quarterly data from 2004Q1 to 2025Q2 (86 quarters). Table 6 lists each observable, its source, and the transformation applied. All raw series are seasonally adjusted at source where applicable; logarithms of levels are taken where indicated. The processed data underlying the estimation is in the replication archive at `brazil_wp/python/data/estimation_data.csv`.

Five observables receive non-zero measurement error in the state-space specification: the world oil and food commodity gaps, the U.S. output gap, the risk premium, and the inflation target. The remaining baseline observables, including the terms-of-trade index, are observed without measurement error. The terms-of-trade level trend and trend-growth state are estimated symmetrically with the other model trends through the SNR-calibrated state-space system described in Section 3.3.

Table 6: Data Sources and Transformations

Observable	**Source**	**Transformation**
Real GDP (`OBS_LGDP`)	IBGE, quarterly SA real GDP	$100 \cdot \log$ of level
Headline CPI (`OBS_DA_LCPI`)	BCB / IBGE, IPCA	Annualised quarterly log-difference
Core CPI (`OBS_DA_LCPIX`)	BCB, equal-weighted core IPCA (EX0, EX3, Trimmed Smoothed, Double Weight, P55)	Annualised quarterly log-difference of cumulated index
Energy CPI (`OBS_DA_LCPI_ENERGY`)	IBGE, IPCA energy subgroup	Annualised quarterly log-difference
Nominal exchange rate (`OBS_DA_LS`)	BCB, USDBRL	Annualised quarterly log-difference
Selic policy rate (`OBS_RS`)	BCB, quarterly average of daily Selic	Annualised, in percentage points
Country risk premium (`OBS_PREM`)	Bloomberg, Brazil 5Y CDS	Quarterly average, in percentage points
Inflation target (`OBS_DA_LCPI_TAR`)	BCB CMN target, interpolated to quarterly	Annualised, in percentage points
Terms of trade (`OBS_TT`)	BCB / FUNCEX, IC-Br USD index	$100 \cdot \log$ of level
World oil price gap (`OBS_LRWOIL_GAP`)	FRED, WTI in USD deflated by U.S. CPI	HP-filtered cyclical component
World food price gap (`OBS_LRWFOOD_GAP`)	FAO/IMF food price index deflated by U.S. CPI	HP-filtered cyclical component
U.S. CPI (`OBS_DA_LCPI_US`)	FRED, CPIAUCSL	Annualised quarterly log-difference
U.S. output gap (`OBS_GDP_GAP_US`)	Holston et al. (2017) update	As reported, in percentage points
U.S. federal funds rate (`OBS_RS_US`)	FRED, FEDFUNDS	Annualised, in percentage points
U.S. neutral real rate (`OBS_RR_BAR_US`)	Holston et al. (2017) update	As reported, in percentage points

J Complete Model Equations

This appendix lists the full system of equations for the Brazilian QPM under the baseline specification (Model C, defined in § 4.4), which includes the terms-of-trade trend-growth coupling on potential output of § 4.2. The system is partitioned into a stationary gap block (§ J.1), a non-stationary trend block (§ J.2), and the measurement equations linking states to observables (§ J.3). Variable names follow the convention used in the model source files of the replication archive; the correspondence with the Greek-symbol notation of the body is given in Table 7. All gap variables are stationary deviations from trend; all trend variables follow the law of motion in § J.2; the state-space link is given by § J.3.

Table 7: Notation map: body symbols vs. appendix source-file names

Body symbol	**Appendix name**	**Description**
$\tilde{y}_t$	GDP_GAP	Output gap
$\tilde{z}_t$	LZ_GAP	Real exchange rate gap
$\tilde{r}_t$	RR_GAP	Real-rate gap
tt_t	TT_GAP	Terms-of-trade gap
i_t	RS	Selic rate (gap component)
π_t^x	DA_LCPIX	Core inflation
π_t	DA_LCPI	Headline inflation
$\tilde{\rho}_t$	PREM_GAP	Risk-premium gap
$q_{r,t}$	QRR	Koyck-filtered real-rate gap
$q_{tt,t}$	QTT	Koyck-filtered ToT gap
$\bar{y}_t$	LGDP_BAR	Potential output (log level)
$\bar{g}_t$	G_TREND	Potential growth
$\bar{g}_t^{tt}$	D_TT_BAR	ToT trend-growth state
$\overline{tt}_t$	TT_BAR	Terms-of-trade level trend
$\bar{r}_t$	RR_BAR	Domestic equilibrium real rate
$\bar{\pi}_t$	PI_TREND	Inflation trend
$\bar{\rho}_t$	PREM_BAR	Risk-premium trend
$\overline{\Delta z}_t$	DLZ_TREND	Trend real depreciation
$\bar{z}_t$	LZ_BAR	REER trend (level)

J.1 Gap Block

Aggregate demand (IS curve).

$$\begin{aligned}\text{GDP_GAP}_t = {} & \beta_1 \,\text{GDP_GAP}_{t-1} + \beta_2 \,\mathbb{E}_t[\text{GDP_GAP}_{t+1}] - \beta_3 \,\text{QRR}_t \\ & + \beta_4 \,\text{LZ_GAP}_t + \beta_5 \,\text{GDP_GAP_US}_t + \phi_{tt}^{IS} \,\text{QTT}_t + \text{RESN_GDP_GAP}_t,\end{aligned} \tag{34}$$

$$\text{QRR}_t = \rho_{QRR} \,\text{QRR}_{t-1} + (1 - \rho_{QRR}) \,\text{RR_GAP}_t, \tag{35}$$

$$\text{QTT}_t = \rho_{QTT} \,\text{QTT}_{t-1} + (1 - \rho_{QTT}) \,\text{TT_GAP}_t, \tag{36}$$

$$\text{RESN_GDP_GAP}_t = \rho_{n^{\tilde{y}}} \,\text{RESN_GDP_GAP}_{t-1} + \varepsilon_t^{\tilde{y}}. \tag{37}$$

Inflation block (core, energy, food, headline).

$$\begin{aligned}\text{DA_LCPIX}_t = {} & \lambda_{1x} \,\mathbb{E}_t[\text{DA_LCPIX}_{t+1}] + (1 - \lambda_{1x}) \,\text{DA_LCPIX}_{t-1} \\ & + \lambda_{2x} \,\text{GDP_GAP}_t + \kappa_z \,\text{QZ}_t + \lambda_{4x} \,\text{LZ_GAP}_t \\ & + \lambda_{5x} \,(\text{LRWOIL_GAP}_t + \text{LZ_GAP}_t) \\ & + \lambda_{6x} \,(\text{LRWFOOD_GAP}_t + \text{LZ_GAP}_t) \\ & + \phi_{tt}^{PC} \,\text{TT_GAP}_t + \varepsilon_t^{\pi^x},\end{aligned} \tag{38}$$

$$\text{QZ}_t = \rho_{QZ} \,\text{QZ}_{t-1} + (1 - \rho_{QZ}) \,4 \,(\text{LZ_GAP}_t - \text{LZ_GAP}_{t-1}), \tag{39}$$

$$\begin{aligned}\text{QZ_E}_t = {} & \rho_{QZ,E} \,\text{QZ_E}_{t-1} \\ & + (1 - \rho_{QZ,E}) \,\big[\text{DA_LS}_t \\ & \quad + 4 \,(\text{LRWOIL_GAP}_t - \text{LRWOIL_GAP}_{t-1})\big],\end{aligned} \tag{40}$$

$$\begin{aligned}\text{QZ_F}_t = {} & \rho_{QZ,F} \,\text{QZ_F}_{t-1} \\ & + (1 - \rho_{QZ,F}) \,\big[\text{DA_LS}_t \\ & \quad + 4 \,(\text{LRWFOOD_GAP}_t - \text{LRWFOOD_GAP}_{t-1})\big],\end{aligned} \tag{41}$$

$$\begin{aligned}\text{DA_LCPI_ENERGY}_t = {} & \lambda_{E1} \,\text{DA_LCPI_ENERGY}_{t-1} + (1 - \lambda_{E1}) \,\text{DA_LCPIX}_t \\ & + \theta_E \,\text{QZ_E}_t + \varepsilon_t^{energy},\end{aligned} \tag{42}$$

$$\begin{aligned}\text{DA_LCPI_FOOD}_t = {} & \lambda_{F1} \,\text{DA_LCPI_FOOD}_{t-1} + (1 - \lambda_{F1}) \,\text{DA_LCPIX}_t \\ & + \theta_F \,\text{QZ_F}_t + \varepsilon_t^{food},\end{aligned} \tag{43}$$

$$\begin{aligned}\text{DA_LCPI}_t = {} & (1 - \lambda_{1s} - \lambda_{2s}) \,\text{DA_LCPIX}_t + \lambda_{1s} \,\text{DA_LCPI_ENERGY}_t \\ & + \lambda_{2s} \,\text{DA_LCPI_FOOD}_t + \varepsilon_t^{\pi}.\end{aligned} \tag{44}$$

Monetary policy.

$$\begin{aligned}
\text{D4A_LCPIX}_t &= \tfrac{1}{4}\big(\text{DA_LCPIX}_t + \text{DA_LCPIX}_{t-1} \\
&\quad + \text{DA_LCPIX}_{t-2} + \text{DA_LCPIX}_{t-3}\big), && (45)\\
\text{RS}_t &= \gamma_1\,\text{RS}_{t-1} + (1-\gamma_1)\big(\text{DA_LCPI}_{t+1} + \gamma_2\,\mathbb{E}_t[\text{D4A_LCPIX}_{t+3}] + \gamma_4\,\text{GDP_GAP}_t\big) \\
&\quad + \text{RESN_RS}_t, && (46)\\
\text{RESN_RS}_t &= \rho_{n^i}\,\text{RESN_RS}_{t-1} + \varepsilon_t^i, && (47)\\
\text{RR_GAP}_t &= \text{RS}_t - \mathbb{E}_t[\text{DA_LCPI}_{t+1}], && (48)\\
\text{RR_GAP_US}_t &= \text{RS_US}_t - \mathbb{E}_t[\text{DA_LCPI_US}_{t+1}]. && (49)
\end{aligned}$$

Exchange rate and external sector.

$$\begin{aligned}
\text{LZ_GAP}_t &= \tfrac{\rho_{ls}}{1+\rho_{ls}}\,\text{LZ_GAP}_{t-1} + \tfrac{1}{1+\rho_{ls}}\,\mathbb{E}_t[\text{LZ_GAP}_{t+1}] \\
&\quad - \tfrac{(1-\rho_{ls})\,\psi_{uip}}{4\,(1+\rho_{ls})}\big(\text{RR_GAP}_t - \text{RR_GAP_US}_t - \text{PREM_GAP}_t\big) + \varepsilon_t^{\tilde{z}}, && (50)\\
\text{DA_LS}_t &= 4\,(\text{LZ_GAP}_t - \text{LZ_GAP}_{t-1}) + \text{DA_LCPI}_t - \text{DA_LCPI_US}_t, && (51)\\
\text{PREM_GAP}_t &= \rho_{\tilde{\rho}}\,\text{PREM_GAP}_{t-1} + (1-\rho_{\tilde{\rho}})\big(-\phi_{tt}^{gap}\,\text{TT_GAP}_t\big) + \varepsilon_t^{\tilde{\rho}}. && (52)
\end{aligned}$$

Terms of trade.

$$\text{TT_GAP}_t = \rho_{tt}\,\text{TT_GAP}_{t-1} + \varepsilon_t^{tt}. \tag{53}$$

Foreign block (US, gap variables).

$$\begin{aligned}
\text{GDP_GAP_US}_t &= \beta_{1,US}\,\text{GDP_GAP_US}_{t-1} - \beta_{2,US}\big(\text{RS_US}_{t-1} - \text{DA_LCPI_US}_t\big) + \varepsilon_t^{\tilde{y}^*}, && (54)\\
\text{DA_LCPI_US}_t &= \rho_{\pi^*}\,\text{DA_LCPI_US}_{t-1} + (1-\rho_{\pi^*})\,\beta_{3,US}\,\text{GDP_GAP_US}_{t-1} + \varepsilon_t^{\pi^*}, && (55)\\
\text{RS_US}_t &= \rho_{US}\,\text{RS_US}_{t-1} \\
&\quad + (1-\rho_{US})\big(\text{DA_LCPI_US}_t + \gamma_{2,US}\,\text{DA_LCPI_US}_{t-1} \\
&\quad\quad + \gamma_{4,US}\,\text{GDP_GAP_US}_{t-1}\big) + \varepsilon_t^{i^*}. && (56)
\end{aligned}$$

World commodity prices.

$$\begin{aligned}
\text{LRWOIL_GAP}_t &= \rho_{oil}\,\text{LRWOIL_GAP}_{t-1} + \varepsilon_t^{oil}, && (57)\\
\text{LRWFOOD_GAP}_t &= \rho_{food}\,\text{LRWFOOD_GAP}_{t-1} + \varepsilon_t^{food,\,gap}. && (58)
\end{aligned}$$

J.2 Trend Block

Level trends (random walk with drift).

$$\text{LGDP_BAR}_t = \text{LGDP_BAR}_{t-1} + \text{G_TREND}_{t-1} + \varepsilon_t^{\bar{y}}, \tag{59}$$

$$\text{LZ_BAR}_t = \text{LZ_BAR}_{t-1} + \text{DLZ_TREND}_{t-1} + \varepsilon_t^{\bar{z}}, \tag{60}$$

$$\text{TT_BAR}_t = \text{TT_BAR}_{t-1} + \text{D_TT_BAR}_{t-1} + \varepsilon_t^{\overline{tt}}. \tag{61}$$

The D_TT_BAR_{t-1} term in the TT_BAR_t law of motion couples the terms-of-trade level trend to its own trend-growth state, replacing the pure random-walk specification used in earlier QPM implementations. The state D_TT_BAR_t — the per-quarter trend growth rate of the terms of trade — is defined among the rate/flow trends below.

Rate / flow trends (mean-reverting to steady state).

$$\text{G_TREND}_t = (1-\alpha_g)\,\text{G_TREND}_{t-1} + \alpha_g\, g_{ss} + \kappa_{g,tt}\left(\text{D_TT_BAR}_{t-1} - d_{tt,ss}\right) + \varepsilon_t^{g}, \tag{62}$$

$$\text{DLZ_TREND}_t = (1-\alpha_{\Delta z})\,\text{DLZ_TREND}_{t-1} + \alpha_{\Delta z}\,\Delta z_{ss} + \varepsilon_t^{\Delta\bar{z}}, \tag{63}$$

$$\text{D_TT_BAR}_t = (1-\alpha_{d,tt})\,\text{D_TT_BAR}_{t-1} + \alpha_{d,tt}\, d_{tt,ss} + \varepsilon_t^{\overline{\Delta tt}}, \tag{64}$$

$$\text{PREM_BAR}_t = (1-\alpha_{prem})\,\text{PREM_BAR}_{t-1} + \alpha_{prem}\,\rho_{ss} + \varepsilon_t^{\bar{\rho}}, \tag{65}$$

$$\begin{aligned}\text{RR_BAR}_t = {} & (1-\alpha_{rr})\,\text{RR_BAR}_{t-1} \\ & + \alpha_{rr}\left(\text{RR_BAR_US}_{t-1} + \text{PREM_BAR}_{t-1} + 4\,\Delta z_{ss}\right) + \varepsilon_t^{\bar{r}},\end{aligned} \tag{66}$$

$$\text{PI_TREND}_t = (1-\alpha_{\pi})\,\text{PI_TREND}_{t-1} + \alpha_{\pi}\,\pi_{ss}^{tar} + \varepsilon_t^{\bar{\pi}}, \tag{67}$$

$$\text{RR_BAR_US}_t = (1-\alpha_{rr^*})\,\text{RR_BAR_US}_{t-1} + \alpha_{rr^*}\, r_{ss}^{*} + \varepsilon_t^{\bar{r}^*}, \tag{68}$$

$$\text{PI_TREND_US}_t = (1-\alpha_{\pi^*})\,\text{PI_TREND_US}_{t-1} + \alpha_{\pi^*}\,\pi_{ss}^{*} + \varepsilon_t^{\bar{\pi}^*}. \tag{69}$$

In the Brazilian application, the loading $\kappa_{g,tt}$ in eq. (62) is estimated under a Gamma prior centred at 0.020 with standard deviation 0.015, following the joint specification of González et al. (2020) (4GM eqs. 2 and 9) and Demidenko et al. (2016, eq. 16) and the identification analysis discussed in § K.2.[6] We fix $d_{tt,ss} = 0$ and the convergence speed $\alpha_{d,tt} = 0.10$.

[6]The 4GM specification also allows a coupling of the ToT trend-growth state into the trend real-exchange-rate depreciation, of the form $\text{DLZ_TREND}_t = (1-\alpha_{\Delta z})\text{DLZ_TREND}_{t-1} + \alpha_{\Delta z}\,\Delta z_{ss} - \nu_{\Delta z,tt}\,(\text{D_TT_BAR}_{t-1} - d_{tt,ss}) + \varepsilon_t^{\Delta\bar{z}}$, with loading $\nu_{\Delta z,tt}$. Identification of this loading is weak in the Brazilian sample, so we set $\nu_{\Delta z,tt} = 0$ and report the trend-depreciation block in eq. (63) without the channel. The posterior under $\nu_{\Delta z,tt}$ estimated freely (results available on request) is essentially identical to the headline.

Setting $\kappa_{g,tt} = 0$ and shutting off the D_TT_BAR shock variance recovers the pure-random-walk specification for TT_BAR_t reported as a robustness comparison in Appendix H.

Contemporaneous trend definitions.

$$\text{RS_TREND}_t = \text{RR_BAR}_t + \text{PI_TREND}_t, \tag{70}$$

$$\text{RS_US_TREND}_t = \text{RR_BAR_US}_t + \text{PI_TREND_US}_t, \tag{71}$$

$$\text{DA_LS_TREND}_t = 4\,\text{DLZ_TREND}_t + \text{PI_TREND}_t - \text{PI_TREND_US}_t. \tag{72}$$

Setting $\alpha_{rr} = 0$ in the RR_BAR_t equation removes the UIP-attractor term and leaves the domestic equilibrium real rate as a no-attractor random walk; this no-attractor neutral-rate restriction is tested in § 5.2.

J.3 Measurement Equations

The model is brought to data through $K = 15$ observables listed below.

Baseline observables.

$$\text{OBS_LGDP}_t = \text{GDP_GAP}_t + \text{LGDP_BAR}_t, \tag{73}$$

$$\text{OBS_DA_LCPI}_t = \text{DA_LCPI}_t + \text{PI_TREND}_t, \tag{74}$$

$$\text{OBS_DA_LCPIX}_t = \text{DA_LCPIX}_t + \text{PI_TREND}_t, \tag{75}$$

$$\text{OBS_DA_LCPI_ENERGY}_t = \text{DA_LCPI_ENERGY}_t + \text{PI_TREND}_t, \tag{76}$$

$$\text{OBS_DA_LCPI_US}_t = \text{DA_LCPI_US}_t + \text{PI_TREND_US}_t, \tag{77}$$

$$\text{OBS_DA_LS}_t = \text{DA_LS}_t + \text{DA_LS_TREND}_t, \tag{78}$$

$$\text{OBS_LRWOIL_GAP}_t = \text{LRWOIL_GAP}_t, \tag{79}$$

$$\text{OBS_LRWFOOD_GAP}_t = \text{LRWFOOD_GAP}_t, \tag{80}$$

$$\text{OBS_GDP_GAP_US}_t = \text{GDP_GAP_US}_t, \tag{81}$$

$$\text{OBS_PREM}_t = \text{PREM_GAP}_t + \text{PREM_BAR}_t, \tag{82}$$

$$\text{OBS_RS}_t = \text{RS}_t + \text{RS_TREND}_t, \tag{83}$$

$$\text{OBS_RS_US}_t = \text{RS_US}_t + \text{RS_US_TREND}_t, \tag{84}$$

$$\text{OBS_TT}_t = \text{TT_GAP}_t + \text{TT_BAR}_t, \tag{85}$$

$$\text{OBS_DA_LCPI_TAR}_t = \text{PI_TREND}_t, \tag{86}$$

$$\text{OBS_RR_BAR_US}_t = \text{RR_BAR_US}_t. \tag{87}$$

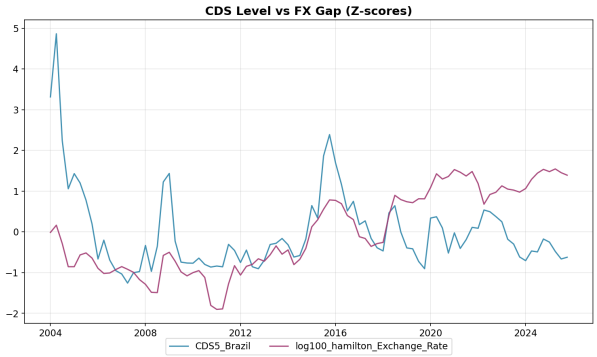
J.4 Shock processes

All structural shocks $\varepsilon_t^{\cdot}$ are independent Gaussian innovations with mean zero and standard deviations $\sigma_{\cdot}$ governed by the prior calibration of § 3; the corresponding SHK_· names in the QPM source file map one-to-one to the symbols used here.

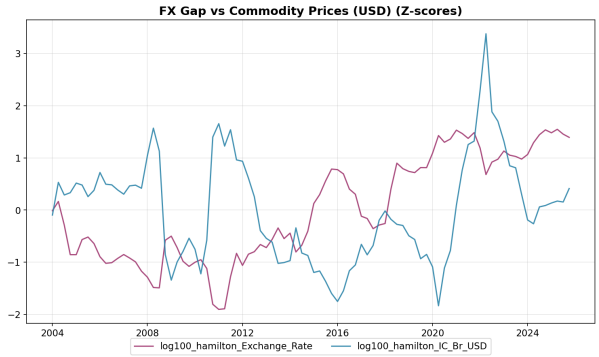
K Additional Empirical Exhibits

K.1 Additional Stylized-Fact Evidence

Figure 17: Exchange Rate Drivers: Risk Premium and Commodity Prices



(a) CDS Level vs. FX Gap ($r = 0.20$)



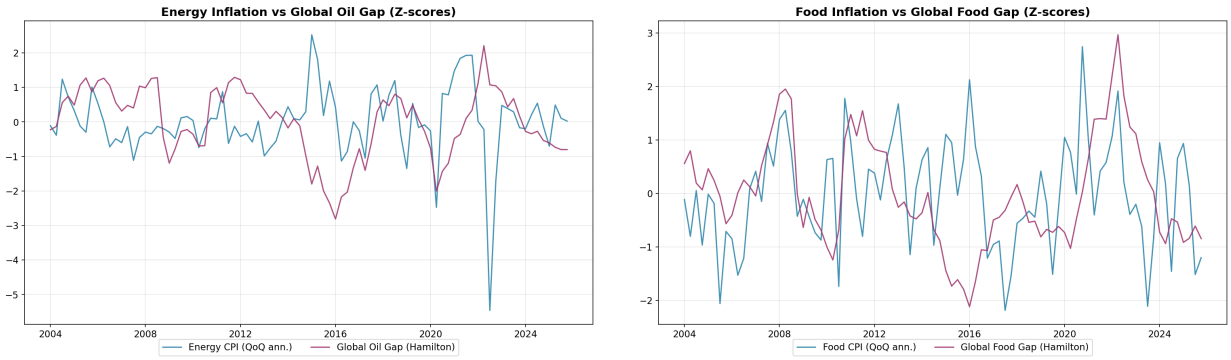
(b) FX Gap vs. Commodity Prices USD ($r = -0.11$)

Note: Hamilton-filtered cyclical components, standardized. Co-movement is most visible during the 2008–09 and 2014–16 episodes.

Exchange rate movements, in turn, pass through to domestic inflation, but through institutionally mediated channels. For energy, Brazil's administered pricing regime (Petrobras fuel pricing, ANEEL electricity tariffs) substantially delays the transmission of global oil shocks, resulting in a weak contemporaneous correlation. For food, where prices are more market-determined, the co-movement is more immediate (Figure 18). This asymmetry motivates the Koyck distributed-lag filters ($\rho_{QZ,E}$ and $\rho_{QZ,F}$) and the disaggregated supply block with separate oil and food price processes (Minella et al., 2003; Camacho and Perez-Quiros, 2013).

Table 8 summarises the mapping from the three stylized facts to the specific structural coefficients of the model presented in § 4.2.

Figure 18: Pass-Through from Global to Domestic Prices



(a) Energy Inflation vs. Global Oil Gap (b) Food Inflation vs. Global Food Gap

Note: Global commodity price gaps (Hamilton-filtered, standardized) vs. corresponding CPI components (annualized QoQ, standardized). The weak energy correlation reflects administered pricing; the stronger food correlation reflects market pricing.

Table 8: Mapping from Stylized Facts to Model Coefficients

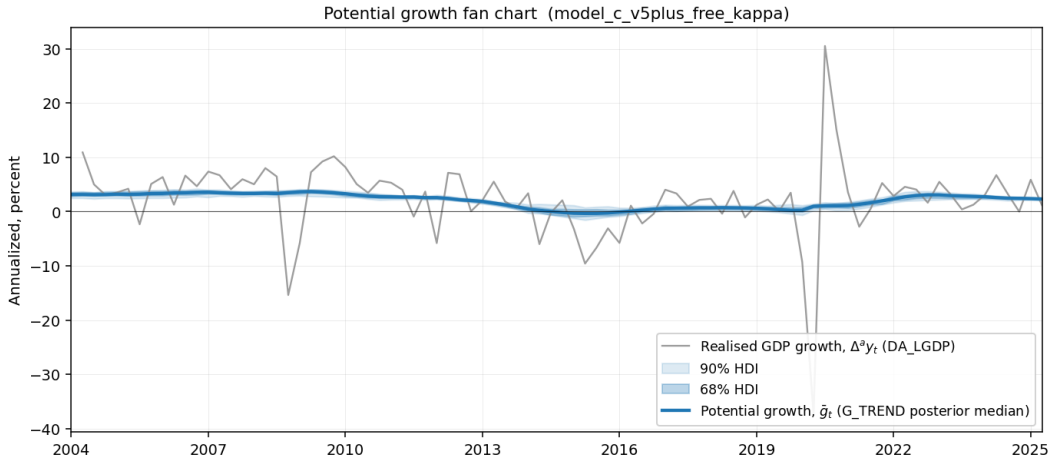
Stylized fact	**Model coefficient(s) / channel**	**Equation**
Commodity-cycle co-movement with the GDP gap (Fact 1)	ϕ_{tt}^{IS}: direct ToT entry in the IS curve; $\kappa_{g,tt}$: ToT-trend coupling on potential output growth.	Eqs. (8) and (13).
Exchange rate absorbs external risk and commodity shocks (Fact 2)	Risk-premium-sensitive UIP attractor; risk-premium-on-ToT loading in the $\tilde{\rho}_t$ equation.	Eq. (11) (and risk-premium block, App. C).
Sluggish energy pass-through vs. market-driven food pass-through (Fact 2)	$\rho_{QZ,E}$ and $\rho_{QZ,F}$: separate Koyck filters on the energy- and food-Phillips curves.	Disaggregated supply block, App. C.
Forward-looking policy and anchored expectations (Fact 3)	$\gamma_1, \gamma_2, \gamma_4$: Taylor rule with smoothing, inflation response and output-gap response; α_π: convergence speed of $\bar{\pi}_t$ to target.	Eq. (10) and trend block.

Note: Each fact maps to one or more structural coefficients or channels; equation labels refer to the full model presented in § 4.2 and Appendix J.

K.2 Potential Growth

Figure 19 reports the time-varying potential growth rate $\bar{g}_t$ (annualised) for the baseline Model C alongside the realised quarterly GDP growth $\Delta^a y_t$. The posterior median declines from approximately 3 percent in the mid-2000s to near zero during the 2014–2016 recession, consistent with the broader literature on Brazil's productivity slowdown, and recovers to roughly 2–3 percent by the end of the sample. The 68% HDI is of order ±0.3 percentage points through the bulk of the sample, widening noticeably during the COVID-19 quarters and reflecting the difficulty of disentangling permanent from transitory output losses in real time. The annualised range of $\bar{g}_t$ across the body specifications under the headline treatment that estimates $\kappa_{g,tt}$ is 6.4 percentage points (Model C), with the alternative body specifications between 5.8 (Model A) and 6.5 (Model B), indicating cross-specification stability of the trend-growth path.

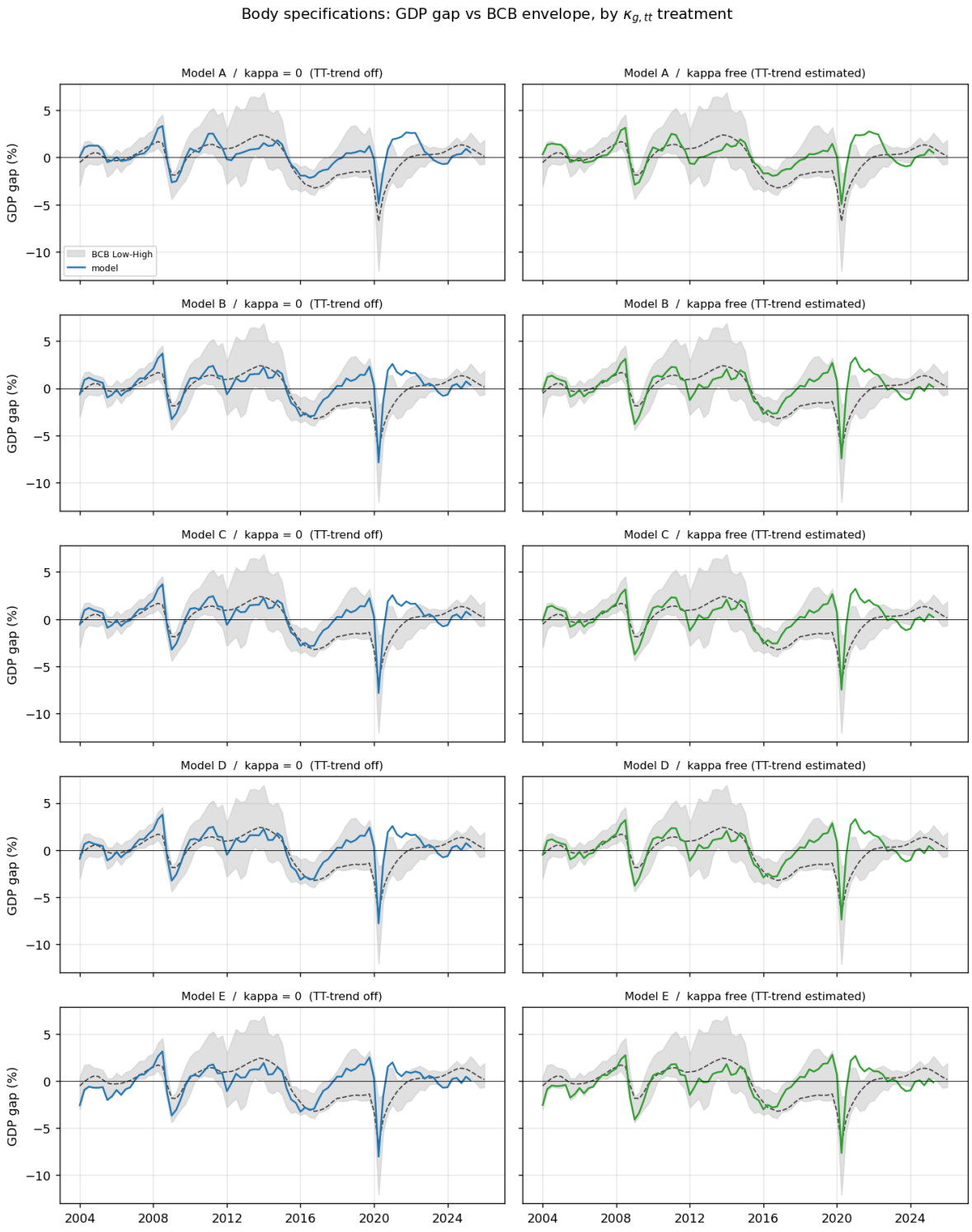
Figure 19: Potential Growth and Realised GDP Growth (Model C)



Note: Posterior median of the trend-growth state $\bar{g}_t$ (annualised) for Model C, with 68% and 90% highest density intervals (shaded), overlaid on the realised quarterly GDP growth $\Delta^a y_t$ (grey line). Sample: 2004Q1–2025Q2.

The post-2022 recovery profile of $\bar{g}_t$ is muted relative to realised quarterly growth: tying potential growth to the persistent terms-of-trade trend through the structural channel of eqs. (13)–(14) attributes a portion of the post-pandemic upward revision in realised growth to the persistent component of the terms-of-trade dynamics rather than to a one-off revision of the cycle. This is consistent with the broader empirical evidence that commodity-cycle dynamics discipline trend growth in commodity-exporting open economies (Fernández et al., 2018; Schmitt-Grohé and Uribe, 2018; Catão et al., 2008).

Figure 20: Cross-Specification Posterior Output Gap, by $\kappa_{g,tt}$ Treatment



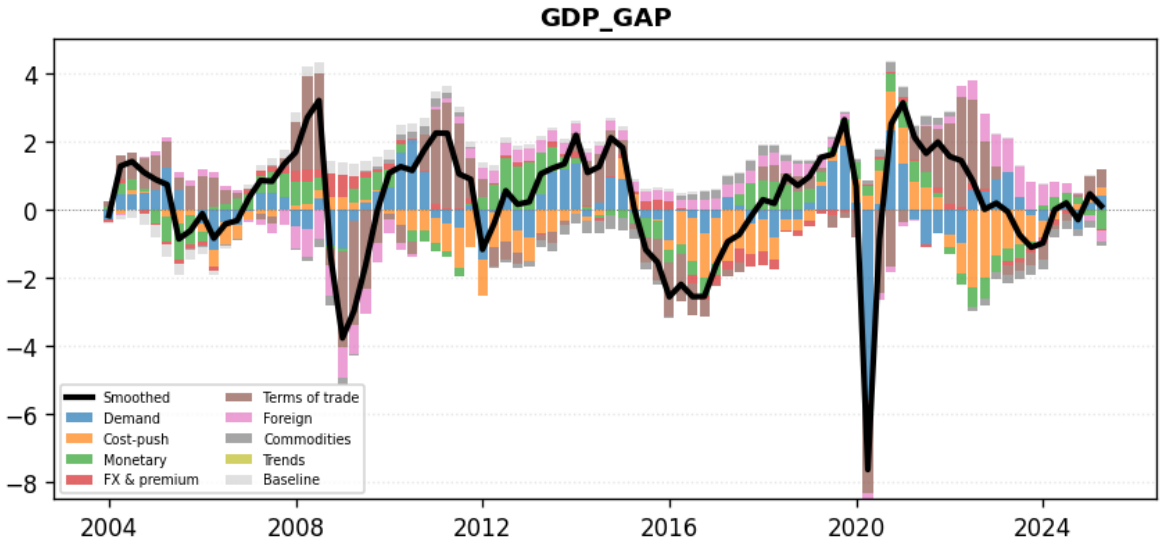
Note: Posterior-median GDP gap for the body specifications (rows) under the two treatments of $\kappa_{g,tt}$ (columns: zero and estimated under Gamma prior). Model E is the COVID-mask robustness reported in Appendix H. Sample: 2004Q1–2025Q2.

The semi-structural QPM thus produces gap estimates that are operationally credible: a model-implied gap reading that aligns with the central bank's own assessment in roughly five of six quarters of the sample is a non-trivial result for a fully estimated framework.

K.3 Historical Shock Decomposition

Figure 21 reports the historical shock decomposition of the Brazilian output gap into its contributing structural shocks. The decomposition reveals four episodes with clear narrative interpretations. In the commodity boom of 2005–2010, positive contributions from terms-of-trade and external-demand shocks dominate. The 2014–16 recession is driven by a combination of adverse terms-of-trade, exchange-rate, and risk-premium shocks. The 2020Q2 trough is, mechanically, dominated by a large negative demand shock, consistent with the global pandemic literature; we do not attempt to attribute this episode to model-internal mechanisms in the body and refer interested readers to the COVID-masked alternative in Appendix H. A single aggregate output gap cannot separately identify demand and supply, so the 2020 shortfall is assigned to the demand residual; a multi-sector or explicit labor-supply shock would be needed to separate them. The 2021–2024 recovery shows positive contributions from terms-of-trade and external-demand shocks.

Figure 21: Historical Shock Decomposition of the Output Gap (Model C)



Note: Stacked bars give the contribution of each structural shock group (demand, supply, terms of trade, exchange rate / risk premium, foreign) to the posterior-median output gap, with the gap itself overlaid (solid line). Sample: 2004Q1–2025Q2.

The substantive role of the terms-of-trade channel in the decomposition is consistent with a long-standing strand of the small-open-economy literature (Fernández et al., 2018; Schmitt-Grohé and Uribe, 2018; Catão et al., 2008): for commodity exporters with flexible exchange rates, terms-of-trade shocks are a dominant driver of the cyclical position, operating through both demand (the IS channel ϕ_{tt}^{IS} in eq. (8)), the risk-premium channel (the negative coefficient on tt_t in the $\tilde{\rho}_t$ equation), and the trend-growth channel of eq. (13). The smoothed states confirm this complementarity directly: the contemporaneous correlation between the cyclical terms-of-trade gap and the GDP gap is +0.80, and the correlation between the cyclical risk-premium gap and the terms-of-trade gap is −0.31 (negative, as the model specifies). Both correlations are stronger than under the pure-random-walk terms-of-trade alternative (+0.75 and −0.19, respectively), an expected consequence of the structural channel: by routing the persistent terms-of-trade variation through the trend block rather than absorbing it into the cyclical residual, more of the cyclical co-movement becomes attributable to genuine ToT propagation.

Historical shock decompositions for the remaining endogenous variables are reported in Appendix F, alongside a discussion of the trend-variable decompositions that, in line with the broader semi-structural-modeling literature, require an interpretive judgement that the body of this paper does not pursue.

K.4 Exchange-Rate Pass-Through

Cumulative pass-through coefficients quantify the impact of nominal exchange-rate and policy-rate shocks on headline inflation over a twelve-quarter horizon and complement the sacrifice ratio in characterising the model's transmission of monetary and external disturbances. Under the baseline Model C, the posterior median twelve-quarter cumulative response of headline CPI inflation π_t to a contractionary monetary policy shock ε_t^i is 1.85 percentage points per percentage point of shock impulse, with 68% HDI [1.53, 2.40]; the corresponding response to a real-exchange-rate-gap shock $\varepsilon_t^{\tilde{z}}$ (an exogenous depreciation) is 1.09 percentage points per percentage point of shock impulse, with 68% HDI [0.90, 1.37].

Across the body specifications and the COVID-mask variant (Models A–E) in the headline treatment that estimates $\kappa_{g,tt}$, the posterior median policy-shock pass-through ranges from 1.44 (Model E) to 2.08 (Model A); the depreciation pass-through ranges from 0.97 (Model A) to 1.38 (Model E). Both ranges lie inside the within-Model-C 68% HDI for the respective coefficient, again confirming that cross-specification variation is small relative to posterior uncertainty.

K.5 Forecast Tables

Table 9: Unconditional Forecast RMSE, Full Sample

Variable	Model	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 8$
	A	3.75	4.43	4.25	4.36	4.18
	B	3.75	4.44	4.25	4.34	4.15
Headline CPI inflation	**C (baseline)**	3.69	4.34	4.17	4.27	4.15
	D	3.72	4.47	4.36	4.51	4.37
	BVAR	6.64	13.63	19.58	11.06	7.15
	A	1.52	2.28	2.42	2.65	3.13
	B	1.53	2.30	2.45	2.67	3.12
Core CPI inflation	**C (baseline)**	1.53	2.30	2.44	2.65	3.10
	D	1.56	2.39	2.58	2.84	3.46
	BVAR	5.51	12.06	4.52	4.22	4.43
	A	10.90	10.80	10.76	10.74	10.77
	B	10.56	10.52	10.73	10.75	10.78
GDP growth	**C (baseline)**	10.54	10.51	10.74	10.76	10.78
	D	10.57	10.51	10.70	10.74	10.78
	BVAR	11.49	24.85	35.29	45.17	24.35
	A	0.54	1.27	2.01	2.72	4.98
	B	0.54	1.28	2.05	2.78	5.08
Selic policy rate	**C (baseline)**	0.54	1.28	2.03	2.75	5.02
	D	0.59	1.35	2.14	2.88	5.25
	BVAR	1.28	3.00	4.42	7.22	15.00

Note: Root mean squared forecast errors in annualised percentage points (annualised quarter-on-quarter inflation, GDP growth and Selic policy rate), computed over 24 forecast origins (2017Q1–2022Q4) and all (origin, horizon) pairs whose target falls inside the sample. QPM specifications use the full-sample posterior median parameter vector; the BVAR is re-estimated at each origin under a Minnesota prior. Baseline (Model C) labelled in bold.

Table 10: Unconditional Forecast RMSE, COVID-Excluded Sample

Variable	**Model**	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 8$
	A	3.41	3.83	3.65	3.37	3.37
	B	3.42	3.83	3.64	3.36	3.31
Headline CPI inflation	**C (baseline)**	3.38	3.79	3.62	3.33	3.32
	D	3.40	3.93	3.81	3.59	3.42
	BVAR	3.85	4.35	4.88	4.82	6.86
	A	1.29	1.87	2.12	2.26	1.87
	B	1.30	1.89	2.14	2.28	1.84
Core CPI inflation	**C (baseline)**	1.30	1.89	2.14	2.27	1.83
	D	1.31	1.93	2.21	2.40	2.08
	BVAR	1.60	2.47	2.84	3.06	2.35
	A	2.72	2.43	2.53	2.29	2.34
	B	2.61	2.42	2.56	2.23	2.40
GDP growth	**C (baseline)**	2.60	2.41	2.55	2.22	2.40
	D	2.50	2.32	2.44	2.15	2.41
	BVAR	4.96	6.01	6.90	6.14	9.57
	A	0.55	1.28	1.98	2.47	3.29
	B	0.56	1.31	2.03	2.56	3.42
Selic policy rate	**C (baseline)**	0.56	1.30	2.01	2.53	3.36
	D	0.62	1.41	2.18	2.74	3.68
	BVAR	0.74	1.78	2.73	3.50	13.39

Note: As Table 9 but excluding forecast observations for which either the origin or target quarter lies in 2020Q1–Q4. After the filter the remaining sample sizes per horizon are $n_{h=1} = 19$, $n_{h=2} = 18$, $n_{h=3} = 17$, $n_{h=4} = 16$ and $n_{h=8} = 16$ (out of the 24 origins).

Table 11: Relative RMSE Reduction from Conditioning, Full Sample (%)

Variable	Model	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 8$
	A	43.06	37.43	33.55	33.28	14.41
	B	42.81	37.40	33.68	34.19	15.54
Headline CPI inflation	**C (baseline)**	41.47	35.98	32.46	32.71	14.82
	D	43.10	37.94	34.87	34.60	7.51
	BVAR	55.60	74.84	83.04	68.77	61.51
	A	2.92	2.92	0.97	-1.33	-7.02
	B	4.55	5.27	4.24	2.53	-4.21
Core CPI inflation	**C (baseline)**	4.23	4.74	3.50	1.68	-4.74
	D	3.15	3.01	0.81	-2.32	-15.66
	BVAR	68.81	79.32	38.58	27.30	36.43
	A	1.92	1.85	1.99	2.14	2.14
	B	4.93	5.23	5.83	5.93	5.91
GDP growth	**C (baseline)**	4.18	4.53	5.11	5.22	5.23
	D	3.73	4.18	4.49	4.46	4.60
	BVAR	14.53	63.69	73.12	78.93	61.02
	A	6.76	8.51	8.56	9.24	7.05
	B	6.36	9.19	10.00	11.27	10.35
Selic policy rate	**C (baseline)**	6.01	8.06	8.47	9.42	8.51
	D	2.15	7.73	8.62	9.12	6.55
	BVAR	5.87	9.11	4.67	24.05	48.68

Note: Relative RMSE reduction from conditioning, in percent: $100 \times (\text{RMSE}_{\text{uncond}} - \text{RMSE}_{\text{cond}}) / \text{RMSE}_{\text{uncond}}$ against the matching unconditional RMSE in Table 9. Each model receives the realised future path of the seven external and exogenous variables (oil and food commodity gaps, U.S. CPI, U.S. output gap, U.S. rates, domestic energy CPI) via the Waggoner and Zha (1999) hard-constraint algorithm and forecasts the remaining seven domestic variables freely. A positive entry means conditioning lowers RMSE; a negative entry means it raises RMSE. Sample as Table 9.

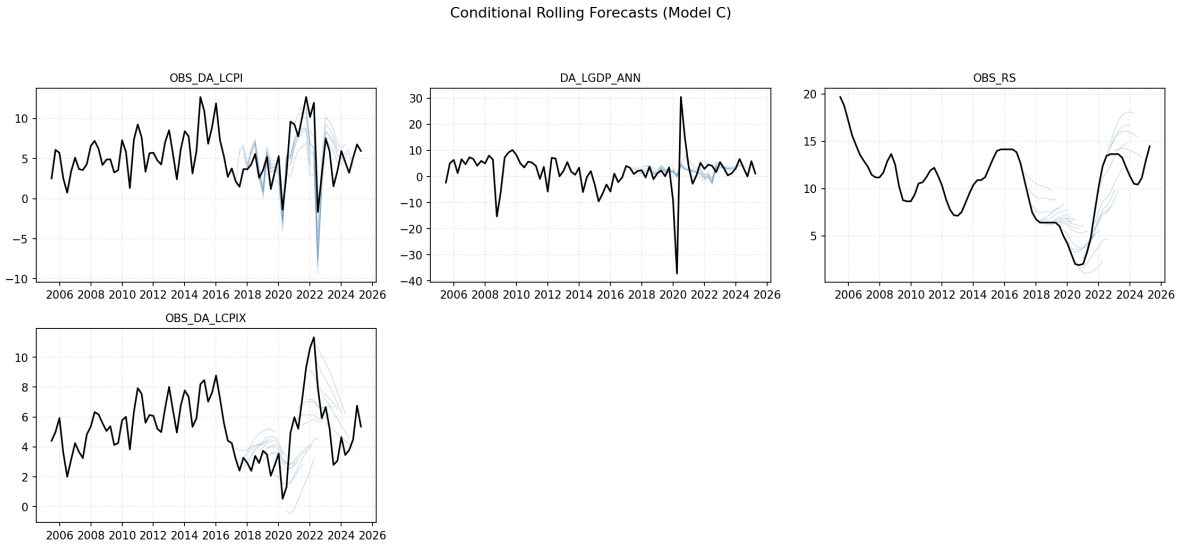
Table 12: Relative RMSE Reduction from Conditioning, COVID-Excluded Sample (%)

Variable	**Model**	$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 8$
	A	43.83	38.63	23.35	7.57	25.59
	B	43.11	38.00	22.90	8.09	25.60
Headline CPI inflation	**C (baseline)**	41.72	36.99	22.04	6.18	26.60
	D	44.54	41.82	27.48	12.38	9.13
	BVAR	20.48	24.02	33.91	28.56	68.71
	A	1.05	-0.01	-5.13	-10.26	-48.66
	B	2.57	2.49	-2.06	-6.84	-46.88
Core CPI inflation	**C (baseline)**	2.35	2.07	-2.61	-7.54	-46.26
	D	0.93	-0.44	-5.90	-12.37	-72.34
	BVAR	-3.13	6.39	7.38	2.12	6.94
	A	-1.68	-3.14	0.20	1.92	2.75
	B	-18.18	-19.88	-10.04	-14.41	10.00
GDP growth	**C (baseline)**	-17.36	-19.53	-9.55	-13.46	10.33
	D	-7.94	-8.36	-2.37	-3.93	9.63
	BVAR	3.38	30.70	22.03	24.58	62.04
	A	3.07	1.69	-2.16	-6.88	-14.01
	B	3.40	2.64	-0.19	-3.61	-7.73
Selic policy rate	**C (baseline)**	2.93	1.74	-1.34	-5.13	-8.43
	D	0.88	2.95	0.75	-3.45	-8.43
	BVAR	-53.34	-36.72	-26.07	-19.35	47.22

Note: As Table 11 (relative RMSE reduction from conditioning, percent) but excluding forecast observations for which either the origin or target quarter lies in 2020Q1–Q4, against the matching unconditional RMSE in Table 10.

K.6 Conditional Rolling Forecasts

Figure 22: Conditional Rolling Forecasts (Model C)



Note: As Figure 26, but with each forecast conditioning on the realised future path of the external observables via the Waggoner and Zha (1999) hard-constraint algorithm.

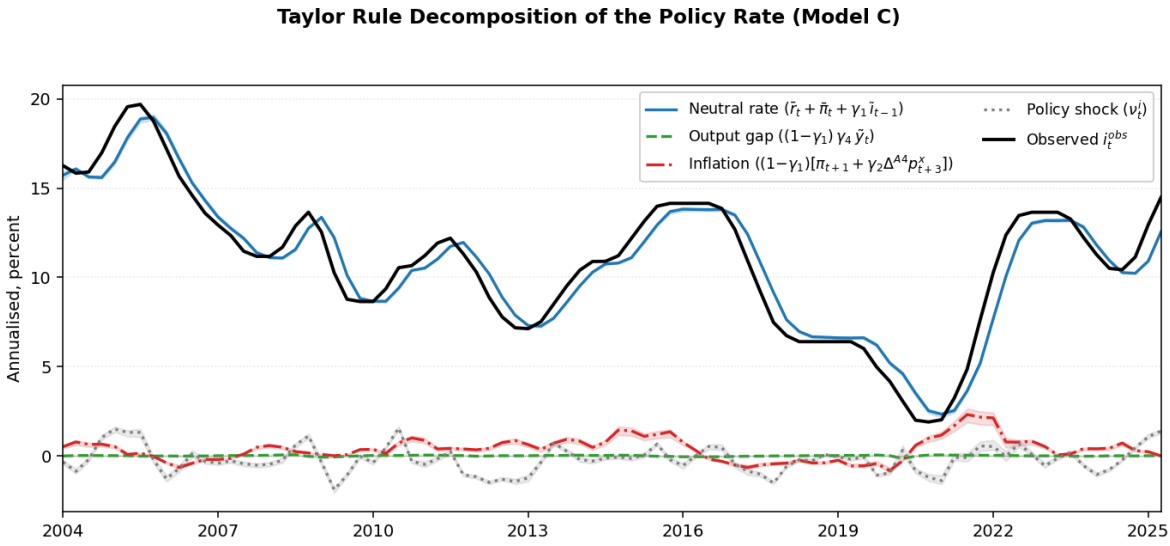
K.7 Additional Tables and Figures

Table 13: Estimated specifications

Specification	**Type**	**Difference from the rich template / baseline**	**Purpose**
Model A	Body	Full distributed-lag structure in the IS curve and pass-through equations	Rich template
Model B	Body	Drops the real-rate Koyck filter in the IS curve	IS parsimony
Model C	Body / baseline	Drops the exchange-rate pass-through Koyck filters in addition to Model B	Headline baseline
Model D	Body	Uses one-quarter-ahead headline inflation in the Taylor rule	Policy timing
Model E	COVID robustness	Masks selected 2020Q1–Q2 observations	Pandemic sensitivity
Model C0	Nested restriction	Sets $\alpha_{rr} = 0$ in Model C, removing the UIP-attractor term for $\bar{r}_t$	Neutral-rate dynamics
RUB1	Prior robustness	Loosens priors on parameters with large prior-to-posterior movements	Prior sensitivity
RUB3	Prior robustness	Weakens the Taylor-rule inflation-response prior	Policy-rule prior
RUB4	Prior robustness	Fixes the inflation-trend convergence parameter	Inflation-trend sensitivity
RUB5	Data robustness	Adds Focus expectations as an inflation-trend observable	Measurement robustness

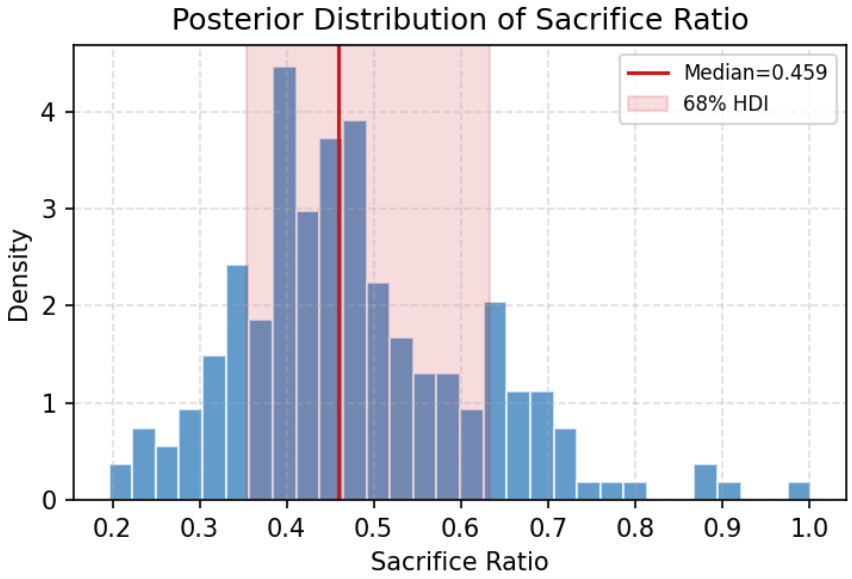
Note: Model C is the baseline. Models A–D vary the structural body; Models E, C0, and RUB1/RUB3/RUB4/RUB5 are robustness or nested-restriction exercises.

Figure 23: Taylor Rule Decomposition of the Policy Rate (Model C)



Note: Additive components of i_t^{obs} from eq. (15). Solid lines are posterior medians; shaded bands are 68% HDI over 200 draws. The thick black line is the observed Selic rate.

Figure 24: Sacrifice Ratio: Posterior Distribution (Model C) with Cross-Specification Region



Note: Posterior distribution from 200 Model C draws. Vertical lines mark the 68% HDI bounds and the cross-specification region of posterior medians. The full table is in Appendix H.

K.8 Stylized Facts and Rolling Forecasts

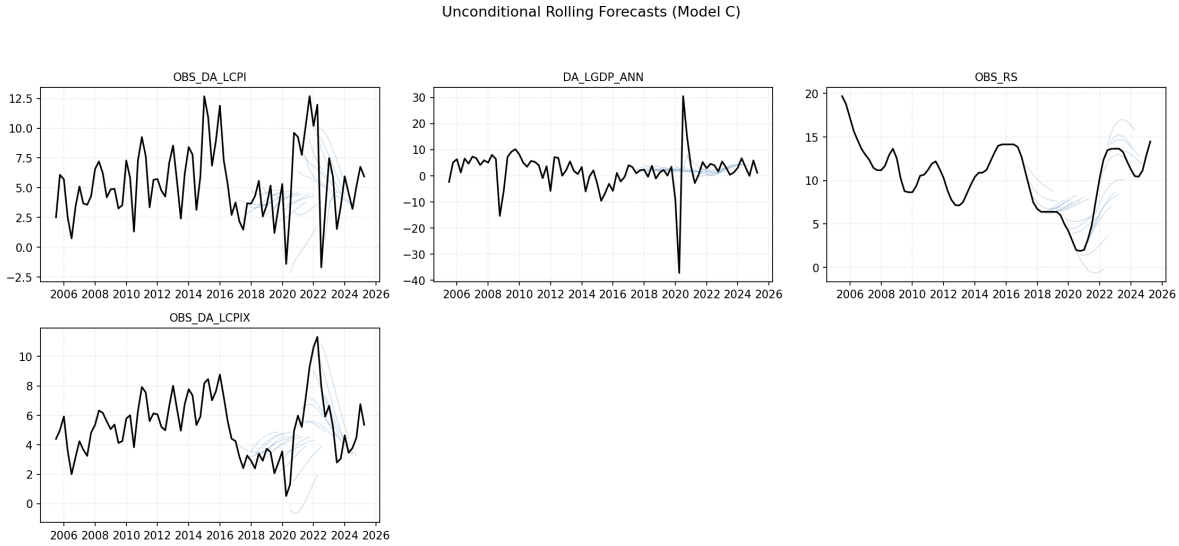
Figure 25: GDP Gap vs. Commodity Prices: The Exchange Rate as Shock Absorber



(a) USD-Denominated ($r = 0.71$) (b) BRL-Denominated ($r = -0.09$)

Note: Hamilton-filtered cyclical components, standardized. The collapse in correlation from USD to BRL quantifies the shock-absorbing role of the flexible exchange rate.

Figure 26: Unconditional Rolling Forecasts (Model C)



Note: One forecast path per origin (2017Q1–2022Q4), out to $h = 8$ quarters. Solid black: realized data. Coloured paths: posterior-median forecasts at successive origins.

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Working Paper No. WP/2026/241